

Chapter 3

Explanatory perspectives

3.1 Methodological issues

Two big questions

*What explains health inequalities? The answer to this question is important for an obvious reason: it determines what can be done to reduce health inequalities. But it may also influence our normative judgement of whether health inequalities are unfair or not: dying younger because of living in poverty will generally be considered unfair, but this is less clear for dying younger because of binge drinking.

There are two big questions with regard to the explanation of health inequalities: (1) do differences in socioeconomic position cause differences in health, or does the 'causal arrow' point from health to socioeconomic position? and (2) what are the specific factors and mechanisms linking socioeconomic position to health (or vice versa)? Finding a scientifically valid answer to these questions poses great methodological challenges, to which we provide an introduction in this paragraph. Readers who wish to skip this technical introduction are of course free to do so, but should be aware of the fact that these methodological difficulties imply that much of our current understanding of the explanation of health inequalities is somewhat provisional.

Causation

Causation versus selection

Starting with the publication of the Black report,²⁰ which brought health inequalities back into the focus of public health research, the question whether 'causation' (i.e. socioeconomic position determining health) or 'selection' (i.e. health determining socioeconomic position, by influencing the selection of people into and out of socioeconomic groups) was the more important mechanism involved in generating health inequalities, has been central to debates about the explanation of socioeconomic inequalities in health.²²⁶ And although some progress has been made, this is true to the present day.

There are several reasons why this issue occupies such a central place. The first is that if socioeconomic position causally determines health, the most effective countermeasures against health inequalities will be providing more years of education to those at the bottom of the social hierarchy, or reducing income inequalities, or other 'egalitarian' social and economic policies. By contrast, if there is no causal relationship between socioeconomic position and health, such policies are unlikely to be effective, at least as far as reducing health inequalities is concerned, although they may still help in achieving other social goals.

* The text of sections 3.1–3.3 has been expanded and adapted from J.P. Mackenbach and J.P. de Jong, 'Health inequalities; an interdisciplinary exploration of socioeconomic position, health and causality'. Copyright © 2018 Federation of European Academies of Medicine (FEAM), All European Academies (ALLEA), and the Royal Netherlands Academy of Arts and Sciences (KNAW). Published under creative commons license licence [Attribution 3.0 Netherlands]: <https://creativecommons.org/licenses/by/3.0/nl/>

A second reason why this issue has been so hotly debated is that **selection mechanisms are often considered to be less of a problem for public policy than causation mechanisms.** Health inequalities due to causation mechanisms are often perceived to be more unfair, particularly by those who feel that the underlying social inequalities are already unfair in themselves.²²⁷ **Scientific disagreements on this issue are therefore to some extent fuelled by ideological differences, with causation explanations being more popular among those leaning towards the political left, and selection explanations being more popular among those with a more conservative or economically liberal outlook.**⁴⁷ (p. 109) However, health-related selection is an equally legitimate object of public policy as causation, as is clear from the fact that many countries have policies in place that reduce the effects of health problems on labour participation and income, and the polarization of this debate has not always been conducive to reaching nuanced conclusions.

Selection explanations stipulate that a person's health determines his or her socioeconomic position, and as this is not less of a causal claim than that of causation explanations, **health-related selection can better be called 'reverse causation.'**ⁱ Like health, socioeconomic position is not a fixed characteristic but changes during a person's lifetime.²²⁸ Because of this 'social mobility' there is, in theory, ample scope for health-related selection processes. For example, health problems may form barriers to achieving a high educational level or to moving up the occupational ladder, and may lead to downward mobility when they experience temporary or permanent loss of paid work. However, whether this mechanism plays an important role in generating health inequalities cannot be determined on theoretical grounds, but needs to be demonstrated empirically.

Confounding

Causation and health-related selection are not the only possible explanations for health inequalities. There is a third possibility, in the form of 'confounding' by 'third factors'.

There are many other factors involved in generating health inequalities than socioeconomic position and health alone, but not all of these should be seen as confounders.²²⁹ (pp. 129–134) **Confounders are third factors which are related to both socioeconomic position and health, but are not on the causal pathway linking socioeconomic position to health.**ⁱⁱ

Confounders should be distinguished from 'mediators.'²³¹ **These are third factors that represent an intermediate step in the causal pathway between socioeconomic position and health.**ⁱⁱⁱ Just as in the case of confounders, mediators are involved in the relationship between socioeconomic position and outcome, but in contrast to **confounders which may not lie on the causal pathway, mediators must lie on this causal pathway.**²²⁹ (p. 186) For example, to the extent that inequalities in **smoking or bad working conditions** are actually determined by people's socioeconomic position, they have to be considered possible **'mediators'**, not **'confounders.'**^{iv}

Nevertheless, there are third factors that can legitimately be considered confounders, because they partly determine people's socioeconomic position, instead of vice versa. For example, the higher prevalence of **obesity** in lower socioeconomic groups is not only due to an effect of socioeconomic disadvantage on the risk of becoming obese, but also to the fact that obese people are discriminated against during recruitment for jobs or promotion.^{132,232} Also, **alcohol-related health problems** may stand in the way of upward occupational mobility, and may even lead to loss of income.²³³ To the extent that the 'causal arrow' points from obesity and alcohol-related health problems to socioeconomic position, their contribution to health inequalities should be considered a form of **confounding**, not mediation.

Potentially more important as confounders are personal characteristics such as **cognitive ability, coping styles, control beliefs, personality, and bodily and mental fitness.** These influence educational and occupational achievement, and at the same time partly determine later health, either directly or indirectly through health-related behaviours such as consumption and exercise

patterns and use of health services.^{234,235,236} To the extent that the ‘causal arrow’ points from these personal characteristics to socioeconomic position, their contribution to health inequalities should be considered a form of **confounding**, not mediation.

! (Social epidemiologists, in recognition of the mechanism through which these personal characteristics are sorted across socioeconomic groups, sometimes use the term ‘indirect selection’ to denote this mechanism, which should be distinguished from the ‘direct selection’ involved when health or health problems affect social mobility.²³⁰

Genetic factors that predispose to ill-health and of which the prevalence varies between socioeconomic groups should also be seen as **potential confounders**, because a person’s genotype temporally precedes his or her socioeconomic position. Again, an association between socioeconomic status and a certain genotype is most likely to arise when that genotype affects social mobility, through an effect on the likelihood of getting a disease that affects social mobility (e.g. mental health problems that stand in the way of educational achievement or upward occupational mobility), through an effect on health-related behaviour that affects social mobility (e.g. predisposition to alcohol addiction), or through an effect on personal characteristics that affect social mobility (e.g. cognitive ability or personality traits).²³⁷

Such selection effects may occur in each new generation, but may also lead to **intergenerational transmission of disadvantage**. Genetic factors predisposing to ill-health that have contributed to the low socioeconomic position of parents may be transmitted to their children, and will make it even more difficult for these children to reach a higher socioeconomic position than their parents. This illustrates that we should not be misled by the somewhat depreciating term ‘confounding’: like ‘health-related selection’, **‘confounding’ can be as policy-relevant as ‘causation’**.

Observation versus experimentation

Because associations between low **socioeconomic position and ill-health may be due to health-related selection** or confounding by third variables, demonstrating that low socioeconomic position causes ill-health requires more than showing that morbidity or mortality rates are higher in lower socioeconomic groups.²³⁸ At the very least, our **study design must enable us to establish that low socioeconomic position precedes ill-health instead of vice versa, and that confounding by other factors can be excluded, either by careful study design** (e.g. by randomization, or by limiting the study to people who are identical in terms of other factors) **or by controlling for these other factors in the analysis**.

In the empirical sciences, the most reliable evidence of causation can be obtained by conducting experiments, but this is mostly unfeasible in the area of socioeconomic inequalities in health. This area therefore relies almost completely on clever observation, in which one tries to approach as closely as possible the ‘clean’ contrast of a controlled experiment. Such a clean contrast is, however, made difficult by the multiple links between socioeconomic position, health and third variables over a person’s lifetime.

Consider the example of two groups of 35-year old and perfectly healthy persons of whom the first has a primary level of education, a routine manual job and an income in the lowest quintile, and the second a tertiary level of education, a professional occupation and an income in the highest quintile. When we follow these two groups over time, we will most likely find a higher incidence of ill-health and premature deaths in the first than in the second group. We have ruled out the possibility of health-related selection because both groups were perfectly healthy at the start of follow-up. But how do we rule out the possibility of confounding by third factors?

One option is to restrict the comparison to 35-year old and perfectly healthy people who are identical in terms of all the personal characteristics that predispose to good or bad health, and that were formed before the individuals had attained their adult socioeconomic position. Candidate factors would include cognitive ability, personality traits, some health-related behaviours already

formed in adolescence, bodily and mental fitness, and genetics, as mentioned above. Another option is to accept a certain degree of heterogeneity of the two groups, and to statistically control for all these potential confounders in the analysis.

However, in the unlikely case that it were practically feasible to select on, or control for, all the confounders, how would we avoid over-adjustment? Over the life-course socioeconomic position and health are linked in a mutually reinforcing way. In our example, removing the effect of third variables that preceded the attainment of their current socioeconomic position of 35-year olds, will also remove some of the effect of socioeconomic conditions in preceding life stages. For example, a person's cognitive ability is not only dependent on his or her genotype, but also on the socioeconomic conditions in which he or she has grown up. In other words, the paradox is that the closer we get to identifying a true causal effect, the farther we may get from a good understanding of how socioeconomic position affects health.

A fundamental divergence of opinion

It is often thought that quasi-experimental studies, which try to mimic randomized experiments by cleverly exploiting 'natural experiments', hold a lot of promise for isolating the causal effects of socioeconomic position on health.^{239,240} Some of these 'quasi-experimental' methods, which have only recently come to be applied in the study of health inequalities, are explained in Box 3.1. While these methods may indeed avoid confounding by both observed and unobserved third variables,²⁴¹ they do not escape the paradox just mentioned.

This paradox actually relates to a fundamental divergence of opinion among scientists on how to approach issues of causality, particularly when it comes to relatively vague concepts such as socioeconomic position. **These quasi-experimental studies are often inspired by the 'counterfactual' approach to causal inference that is increasingly popular in many empirical disciplines, including epidemiology and economics. This uses the 'potential outcomes' framework for assessing causality, which requires that putative causes can be manipulated in experiments, at least in theory.**²⁴⁵

In the case of smoking, it is not difficult to imagine the counterfactual of an otherwise identical person who has never smoked, but in the case of social exposures like low socioeconomic position (or gender, or race), this is much more difficult. **Adherents of the counterfactual approach have therefore argued that assigning causal status to non-manipulable attributes like socioeconomic position (or gender, or race) is problematic, and that only more specifically defined and manipulable exposures qualify as possible causes of health problems.**²⁴⁶

For the study of health inequalities, this point of view implies that it is not useful to ask 'does low socioeconomic position cause ill-health?' For adherents of the **counterfactual approach** this is an unanswerable question that should be replaced by more specific questions, such as **'does an extra year of education cause better health?' or 'does raising the social assistance benefit level cause better health?'** It is the latter type of questions that—in their view—can be answered scientifically, in quasi-experimental (or even truly experimental) studies, and that is also more directly policy relevant.²⁴⁷

However, other epistemological positions have been defended as well. A more pluralistic approach to causality recognizes the **evident existence of causes of ill-health that cannot be manipulated, such as earthquakes, non-white race, or low socioeconomic position.** Although the effects of these exposures cannot be studied according to the rigorous framework prescribed by the counterfactual approach, their role as possible causes of ill-health is considered too important to be ignored.^{248,249} In this alternative view it is precisely because low socioeconomic position often implies life-long exposures which are inextricably linked to social processes—and thus cannot be construed as a counterfactual cause—that it has such pervasive and persistent health effects.

British sociologist John Goldthorpe has argued that the counterfactual approach to causal inference has limited applicability to the central concerns of sociology as a population science. In

UM TIPO DE EXPERIMENTO =
CONTROLOAR OS FATORES PELOS
CONTROLOS → fumo x não fumo

problemático
decidir o que é
causalidade do curso

Box 3.1 Assessing causality between socioeconomic position and health

Schematically, the following design options could be considered to assess whether there is a causal relationship between socioeconomic position and health:^{238,241}

Purely observational studies (i.e. the investigators exploit naturally occurring variations in exposure):

- ◆ cross-sectional and case control studies, for example, in which current health is related to current or previous socioeconomic position, with regression adjustment for observed confounders
- ◆ longitudinal studies, for example, cohort or panel studies relating socioeconomic position to health at a later point in time, with regression adjustment for observed confounders, or fixed effects adjustment for unobserved confounders
- ◆ twin studies, relating socioeconomic position to health of twins who are discordant on socioeconomic position, thereby controlling for genetic factors

Quasi-experimental studies (i.e. investigators exploit ‘natural experiments’ that have created quasi-random variations in exposure):

- ◆ cross-sectional or longitudinal studies with statistical creation of quasi-randomness in the allocation of socioeconomic resources, for example, by propensity score matching, differences-in-differences, instrumental variables, regression discontinuity, ...
- ◆ cross-sectional or longitudinal studies exploiting naturally occurring randomness in the allocation of socioeconomic resources, for example, lotteries, random roll-out of intervention programs, ...

Purely observational studies potentially suffer from many of the problems mentioned in the main text, although (partial) solutions have been proposed. For example, fixed effects adjustment (e.g. analyses which link changes in socioeconomic position to changes in health within the same individual) can be used to remove bias by both observed and unobserved confounders. **Also, studying twins who have ended up in different socioeconomic positions guarantees extensive control for genetic factors.**

Quasi-experimental studies are potentially more powerful in avoiding these problems. For example, in a regression discontinuity analysis one can exploit income thresholds in the allocation of financial benefits to compare health outcomes among people falling just below or above the threshold and therefore getting or just not getting the benefit. Because people just below and just above the threshold are likely to be otherwise similar, this may produce an unbiased estimate of the effect of the financial benefit.²⁴² However, getting close to a clean contrast carries a price: the external validity of the results of regression discontinuity studies for the wider problem of income-related inequalities in health is dependent on whether the health effect of the small difference in financial benefits around the threshold correctly represents the health effects of income along the whole income ladder.

Sometimes, socioeconomic resources are allocated at random in real life. The prime example is lotteries which at first sight offer an excellent opportunity to evaluate the causal effects of money on health—but whether the health impact of an incidental amount of money

Box 3.1 Assessing causality between socioeconomic position and health (*continued*)

obtained in a lottery adequately represents the health effects of years of living on a higher income is unclear.²⁴³ Another example is the roll-out of certain intervention programmes which sometimes occurs more or less at random, as in the roll-out of a higher age of compulsory education across Swedish municipalities in the 1930s.²⁴⁴

Box 3.1 has been reproduced from J.P. Mackenbach and J.P. de Jong, 'Health inequalities; an interdisciplinary exploration of socioeconomic position, health and causality.' Copyright © 2018 Federation of European Academies of Medicine (FEAM), All European Academies (ALLEA), and the Royal Netherlands Academy of Arts and Sciences (KNAW). Published under creative commons licence [Attribution 3.0 Netherlands]: <https://creativecommons.org/licenses/by/3.0/nl/>

his view, sociology primarily aims to explain observed regularities at the population level (such as in our case, health inequalities), and the counterfactual approach addresses different and much more restricted questions. Also, the counterfactual approach does not provide any insight into the underlying mechanisms, and cannot take into account the voluntary nature of human activity, and the fact that human beings are not passive recipients of manipulated exposures. **In Goldthorpe's view, a more fruitful approach is elucidating the mechanisms underlying observed regularities at the population level.**²⁵⁰

In this book we will try to follow a middle road between the two positions, by acknowledging both the strengths and limitations of the counterfactual approach to assessing a causal effect of socioeconomic position (or its more specific components) on health.

Mediation

Mediators

A scientific approach to the explanation of health inequalities cannot stop at the demonstration of an effect of socioeconomic position on health, but also requires an understanding of the factors and mechanisms involved in generating this effect. **We need to be able to identify plausible causal pathways before we can reasonably conclude that socioeconomic position has an effect on health.**²⁵¹

This is, of course, also important from a policy perspective. Broadly speaking, one can distinguish two strategies for reducing health inequalities.²⁵² The first and most radical option is equalizing the distribution of socioeconomic factors, for example by reducing inequalities in educational attainment, employment, or income. To the extent that there is a causal effect of socioeconomic factors on health, this can be expected to also reduce health inequalities.

A second, more pragmatic strategy for reducing health inequalities is to equalize (by 'leveling up') the distribution of specific health determinants across socioeconomic groups.²⁵³ For example, to the extent that socioeconomic inequalities in mortality are determined by differences in working conditions or access to health care, reducing these differences by improving working conditions and access to health care for lower socioeconomic groups can be expected to also reduce health inequalities.^v

The **most commonly used analytical technique for identifying the specific health determinants involved in generating health inequalities is 'mediation analysis'.** This technique allows one to quantify the contribution of one or more 'mediators' to the effect of an **independent variable** (in this case: socioeconomic position) **and a dependent variable** (in this case: a health outcome).

Mediation analysis is based on the notion that health inequalities can be explained by differences in exposure to health determinants between people in lower and higher socioeconomic

pode - se mudar o status quo tanto ↑ salário = CAUSAL como

↓ ≠ DETERMINANTES SAÚDE = HEALTH SELECTION

groups. This is the type of explanation we know most about, and will therefore be the focus of this chapter. However, third variables can not only act as mediators, but also as ‘moderators’ of the relationship between socioeconomic position and health.^{231,254} **A moderator is defined as a variable that affects the strength of the relationship between an independent variable and a health outcome.**^{vi}

This may give rise to ‘differential susceptibility’ or ‘differential vulnerability’, that is, health inequalities may partly be explained by the fact that **people in lower socioeconomic groups are more ‘susceptible’ or ‘vulnerable’ to the negative health effects of various determinants, due to a range of other biological, psychological, and social factors.**^{255,256} Although empirical evidence on such moderation effects is still scarce, there is evidence that it may play a more important role than previously recognized.

For example, it has been shown that excessive alcohol consumption leads to more alcohol-related hospitalizations and deaths in lower socioeconomic groups, perhaps because drinkers from higher socioeconomic groups are protected by better dietary habits, a safer drinking environment, or more support from work and family to address emerging alcohol problems.^{257,258}

Similarly, some studies have found that lower educated smokers have a greater likelihood of developing lung cancer than smokers with a higher level of education. This may be due to subtle differences in smoking behaviour (e.g. deeper inhalation of tobacco smoke or more carcinogenic types of tobacco smoked), but also to differences in biological susceptibility. In either case the contribution of smoking to inequalities in lung cancer will be larger than estimated on the basis of a simple mediation analysis.²⁵⁹

How can mediation be demonstrated?

One approach to assessing mediation is to take the prevalence rates of the health determinant of interest in each socioeconomic group, extract the relative risk of the determinant on morbidity or mortality from the literature, and then estimate the contribution of the determinant to morbidity or mortality inequalities using the method of **population-attributable fractions.**²⁶⁰

Because **data requirements for this approach are modest,** it can be applied in many settings, and has produced estimates of the contribution of various risk factors to inequalities in mortality for many European countries.^{261–264} For example, it suggests that the contribution of smoking to inequalities in mortality ranges between 4% and 26% among men, and between 1% and 20% among women.²⁶²

This is, however, a rather crude approach, and a more accurate assessment of the contribution of **health determinants to health inequalities requires a multivariate analysis of the three-way relationships between socioeconomic position, health determinant and health outcome.** This is what ‘mediation analysis’ does.

The practice of mediation analysis in social epidemiology (and in other disciplines, such as psychology and sociology) has long been based on the so-called Baron and Kenny approach which was developed in the 1980s.^{231,265} Current insights into the contribution of health determinants to health inequalities therefore largely derive from this approach, in which one studies whether the statistical relationship between socioeconomic position and health disappears, either completely or partly, upon controlling for the mediator(s).

This usually takes the form of the so-called ‘difference method’, in which **one estimates the difference between the ‘effect’ of socioeconomic position on a health outcome before and after controlling for the mediator(s).**^{231,265} **This reduction in the size of the ‘effect’ of the independent variable on the health outcome is called ‘attenuation.’**^{vii}

For example, in a cohort study in the Netherlands it was found that the Rate Ratio of mortality among low as compared to high educated men decreased from 1.84 to 1.66 upon controlling for

the higher prevalence of smoking among the low educated, suggesting that the contribution of smoking to educational inequalities in mortality is 21% in this setting (calculated as $100 \times (1.84 - 1.66) / (1.84 - 1.00)$).²⁶⁸

Recently, the 'difference method' has been criticized for a number of methodological shortcomings.^{254,265,269} More specifically, it has been shown that it often gives biased results.^{viii} New methods of mediation analysis have therefore been designed that remove some of these sources of bias.^{254,265} Social epidemiologists have started to apply the new techniques, and found that these may lead to substantially different conclusions as compared to the conventional 'difference' method.²⁷⁰⁻²⁷²

Apart from technical difficulties, a more fundamental shortcoming of mediation analysis is that the results can only be interpreted as evidence for mediation *sensu stricto*, if we assume two causal relationships: socioeconomic position causes exposure to the health determinant, and exposure to the health determinant causes ill-health. Unfortunately, most datasets to which mediation analysis is applied, do not allow us to test that assumption.

In the example of the cohort study on the role of smoking in generating health inequalities in the Netherlands, despite its longitudinal design and adjustment for a range of confounders, due to the possibility of unobserved confounders there is no guarantee that the higher prevalence of smoking among the low educated is caused entirely by their lower socioeconomic position, nor is there a guarantee that the higher mortality among smokers found in the dataset is caused entirely by smoking.²⁶⁸

In order to acknowledge all these uncertainties we will therefore refrain from using the term 'mediator' and use the term 'contributing factor' instead, and we will also largely refrain from presenting quantitative estimates of the role of differences in exposure to health determinants, and instead present our conclusions on the role of contributing factors in more qualitative terms.

3.2 Education, occupation, income, and health

Education and health

From a sociological point of view, socioeconomic position is a multi-faceted phenomenon, which cannot be completely captured by measures of education, occupational class, and income. Nevertheless, the best evidence that we have on the causal effects of low socioeconomic position on health comes from quasi-experimental studies in which socioeconomic position has been disaggregated in its component parts. In this paragraph we review the scientific evidence on the health effects of education, occupational class and income.

Causation, reverse causation, confounding

Many longitudinal studies have shown that adults with a lower level of education have a higher likelihood of falling ill or dying prematurely: indeed, these inequalities have been found in all countries which take the trouble to collect the information.²⁷³ Because most of these health problems arise long after the age at which people complete their education, health-related selection is unlikely to be involved,⁵¹ although some health-related selection may have occurred in a previous life-stage, as children with chronic diseases are somewhat less likely to achieve a higher level of education.²⁷⁴

The main question is whether and, if so to what extent, the association between education and health is confounded by third variables. The short answer is 'yes, it is likely to be confounded, but we do not know by how much,' and confounding is certainly not the only explanation, because there is good evidence from quasi-experimental studies that there is also a causal effect of education on health.

Although variations in educational achievement are partly dependent on parental socioeconomic position, educational achievement is also strongly dependent on a person's own cognitive ability during childhood and adolescence.²⁷⁵ And although children's cognitive ability is partly dependent on the environment in which they grow up, variations in cognitive ability among children are also strongly genetically determined.²⁷⁶ We explain some of the concepts and methods of population genetics in Box 3.2.

It is therefore very likely that the association between education and health is partly confounded by genetic factors. Empirical evidence comes from molecular studies which have recently identified many genetic variants related to education, with 'polygenic scores' now explaining up to 10% of all inter-individual differences in educational achievement.^{284–287} These associations do not necessarily arise from a causal effect of children's genotype on their own educational achievement, but could theoretically also arise from an effect of their parents' genotype (which is, of course, partly similar to that of their children) on how they are raised. However, there is evidence that 'polygenic scores' predicting education also predict upward educational (and occupational class) mobility of children as compared to their parents, suggesting a truly causal effect.²⁸⁸

The pathways linking genotype to education are likely to include, first of all, cognitive ability. We will more extensively review the evidence on the genetic determinants of cognitive ability in section 3.3, 'Six groups of contributing factors'. The pathways are also likely to include genetically determined aspects of personality such as the Big Five personality traits ('openness to experience', 'conscientiousness', 'extraversion', 'agreeableness', and 'neuroticism'), or the personality traits studied in behavioural economics ('self-control', 'risk aversion', and 'time preferences').^{285,289} Because cognitive ability and other (partly) genetically determined personal characteristics are independent predictors of health in later life, one can reasonably infer that genetic factors must to some (as yet unknown) extent confound the relationship between education and health.

Furthermore, the relative contribution of genetic factors to differences in educational achievement has increased over time, due to a decreasing influence of environmental barriers. This is shown, for example, by the fact that the heritability of educational achievement is larger among twins born more recently, after the introduction of educational policies which decreased the influence of family background.^{290,291} Although the evidence on trends is not entirely consistent,^{277,292} between-country comparisons show that the heritability of educational achievement is higher in countries where the environment puts less constraints on educational achievement, such as the Nordic countries.²⁹³

Quasi-experimental evidence

In the presence of these and other risks of confounding, which are difficult to control in observational studies, experimental and quasi-experimental studies may provide more reliable evidence for a causal effect of education on health. Truly experimental evidence is limited to a few studies from the US that have assessed the long-term health effects of early childhood (or pre-school) education. These showed that children receiving preschool education were healthier and less likely to be smoking or obese as adults were.²⁹⁴

The effect of school education on health in later life has repeatedly been assessed in a quasi-experimental set-up. The most common approach has been the evaluation of the impact of compulsory schooling laws.^{238,294–296} During the twentieth century, many countries have introduced such laws, and thereby increased the minimum school leaving age. Because the resulting changes in years of schooling can be regarded as 'exogenous' (i.e. independent of personal attributes of the children involved), any improvements in health occurring in cohorts that left school after the change can reasonably be attributed to the extra years of schooling.

Box 3.2 Genetic analysis: selected concepts and methods

Heritability

Heritability is defined as the proportion of all variation in a ‘phenotypic trait’ (i.e. observed characteristic, such as cognitive ability or health) between individuals in a population that is due to genetic variation between individuals in that population. The total variation in a ‘phenotypic trait’ is thought to consist of three components: genetically induced variation, environmentally induced variation, and random error. This implies that the measured degree of heritability of a ‘phenotypic trait’ depends on the degree of both genetic and environmental variation in a population. In a population in which the environment is more homogeneous, heritability will be higher, and vice versa.²⁷⁷

Twin studies

The classical approach to estimating heritability in human populations is by comparing resemblances between dizygotic and monozygotic twins. Other but conceptually similar approaches involve comparisons between adoptive and biological siblings and between parents and their adoptive and biological offspring. Dizygotic twins on average share half their genes, whereas monozygotic twins share all their genes and therefore are twice as genetically similar as dizygotic twins. The heritability of a particular ‘phenotypic trait’ can thus—under certain assumptions, such as the environmental similarity between the two types of twins being the same—be estimated as approximately twice the difference in correlation between monozygotic and dizygotic twins.²⁷⁷

The heritability gap

Estimates of the heritability of cognitive ability calculated from twin and similar studies range between 30% and 80%, and increases from a value of about 30% in early childhood to much higher values in adulthood and old age.²⁷⁸ Genome-Wide Association Studies (GWAS) have identified many specific genetic variants that are associated with cognitive ability,^{278,279} but there still is a big ‘heritability gap’, that is, the combined effects of all genetic variants that have until now been identified cannot fully account for the heritability estimated from twin studies. This may either be due to violations of the assumptions underlying estimates from twin studies, or be a matter of slow advances in filling the gap as the findings of more and more molecular studies accumulate.^{277,280}

Genome-wide complex trait analysis

Recently a new technique, genome-wide complex trait analysis (GCTA), has been proposed which does not require the often tenuous assumptions of twin studies and other studies based on known family relationships. In this technique, the degree of genetic similarity between unrelated individuals in the population is determined directly on the basis of an analysis of their DNA, using the presence of common single-nucleotide polymorphisms.²⁸¹ Applications of this method have grown explosively, and have produced new estimates for the heritability of many ‘phenotypic traits’ including level of education, socioeconomic position, cognitive ability, personality traits, smoking, alcohol consumption, many chronic diseases, etc. The estimates are often somewhat lower than those from twin studies, but almost always substantial.

Box 3.2 Genetic analysis: selected concepts and methods (continued)

Although promising, it is still a new technique and there is no consensus yet about its relative merits.²⁸²

Mendelian randomization

This approach uses the genetic determinants of a particular 'phenotype' (e.g. a high serum cholesterol, or a low level of education) to study the causal effects of that phenotype on health. The idea behind this analytical strategy is that whether people have certain genes or not is the outcome of a natural experiment occurring at conception, when each child inherits half of each of their parents' genomes. Because whether a child inherits a particular gene from its father or mother is the outcome of a random process, the association between that part of the variation in phenotype (high serum cholesterol, low education) induced by the gene and a health outcome is considered to be less subject to confounding than the association between most other variations in the phenotype and a health outcome.⁵⁶ Application of this idea is, however, challenging, for example, because children's genotypes are only randomly assigned conditional on their parents' genotype, and the latter therefore needs to be controlled.²⁸³

Most of these studies found that longer schooling led to a reduction in mortality in mid-life and beyond, with large variations in effect size.²⁹⁴ Education also reduces the risk of taking up smoking,^{294,295} and better-educated parents have healthier children.²⁹⁵ Although studies exploiting compulsory schooling laws have important limitations (e.g. it is unclear whether the effect of one year of extra schooling at the age of, say, 16 can be generalized to the whole range of variation in length of education currently seen), these findings clearly suggest that there is a causal effect of education on mortality, and that the association between the two seen in observational studies is not only due to confounding by genetic or other usually unobserved factors.

This conclusion is strengthened by the results of recent studies using **Mendelian randomization, a technique exploiting the natural experiment occurring at conception when genes from the mother and father are allocated at random to the child. Using genetic determinants of education as an 'instrumental variable' to mimic randomly occurring variation in educational achievement, these studies found clear evidence for a protective effect of more years of education on dementia,⁵⁷ ischemic heart disease,⁵⁸ and mortality.⁶⁰**

Occupational class and health

As in the case of education, many studies have shown that a lower occupational class is associated with higher rates of morbidity and mortality.^{104,105,297} The main issue is whether this is due to a causal effect of occupational class on health, or due to 'reverse causation' or confounding. Perhaps because of the difficulties involved, no studies with rigorous identification strategies to isolate a causal effect of occupational class on health have been conducted (although there is reasonably strong evidence of the effect of specific working conditions on ill-health).²⁹⁸

Because educational achievement usually precedes entry to the labour market, and because a higher level of education is a requirement for entry into higher occupations, education should be considered a potential confounder of the association between occupational class and ill-health. Studies from the US have sometimes found that the association between occupational class and health disappears after controlling for level of education,^{299,300} but studies from several European countries often show independent (although attenuated) effects.^{51,301,302}

A big difference between education and occupational class is that whereas one's level of education remains constant after the age of, say, 25, one's employment status and occupational class can still change after entering one's first job. This implies that the scope for reverse causation by health-related conditions is much larger in the case of occupational class than in the case of education. A rigorous analysis of the labour market effects of health-related conditions in a range of high-income countries has indeed shown that having a chronic disease, and being a smoker or obese, have negative effects on employment, wages, sick leave, and early retirement³⁰³.

That health-related selection in and out of employment, and during occupational careers does occur is thus undisputed, but there is no consensus on whether these selection effects widen or narrow health inequalities. Several studies have found that the health of people who move downward is worse than that of those who remain in their class of origin, and better than that of those in their class of destination, whereas the health of those who move upward is better than that of others in their class of origin, and worse than that of others in their class of destination. On the basis of this it has been claimed that health-related occupational mobility will tend to 'constrain' or 'dilute' health inequalities.^{304, 305}

While this may seem straightforward, others have demonstrated that this is not necessarily true. **Even if the health of the socially mobile is in-between that of their classes of origin and destination, the net effect of health-related selection may be to widen health inequalities.** This is because the effect of health-related selection does not depend on the difference in health between the socially mobile and *all those in the class they join*, but on the difference in health between the socially mobile and those *whom they replace* in the class they join.^{306,307}

If the health of those who move upward into a higher class is better than that of those who move out of this class, or if the health of those who move downward into a lower class is worse than that of those who move out of this class, health inequalities will widen, not narrow. Detailed studies of the relative numbers and health status of incomers and out-goers to each occupational class have confirmed that health-related selection during intragenerational mobility may indeed widen health inequalities in the population as a whole.^{306,307}

However, whatever the direction of the effect is, the contribution of health-related selection to the explanation of occupational class inequalities in health at adult and higher ages is likely to be limited, because most health problems occur in late middle or old age, after people have reached their final occupational class. This reasoning is confirmed by the fact that longitudinal studies in which occupational class has been measured before health problems are present, and in which the incidence of health problems has been measured during long-term follow-up, also show clearly higher risks of health problems in the lower occupational classes.^{297,308,309}

Income and health

Many studies have also found a **positive association between income and health: people with a higher income tend to have better health and live longer.**^{238,295,310} **The relationship is non-linear: at the lower end of the income distribution, the relationship is steeper than at the upper end, suggesting that whatever mechanisms explain these inequalities, their effects are stronger among those living on a very low income.**^{62,311} Studies that have assessed whether the association between income and health still holds after controlling for education and/or occupational class have often (but not always) found that this is indeed the case^{51,301,312,313}

Causation, reverse causation, confounding

As in the case of education and occupational class, an important question is whether this association is due to a higher income leading to better health (causation), to better health leading to

a higher income (reverse causation), or to confounding by third variables. The common view among public health scientists is that causation accounts for a substantial part of this relationship, whereas the dominant view in the economics literature is that reverse causation is far more important.³¹⁴

Both directions of causality are certainly plausible. A higher level of income may produce better health through several mechanisms. For example, it increases access to healthy foods and good housing conditions, it reduces the stress of financial insecurity and boosts self-confidence, and it removes financial barriers to health care. But better health may also produce a higher income, for example by increasing capacity to work, labour productivity, and wages.³¹⁵ As compared to health inequalities by education and occupational class, health inequalities by income probably have the largest scope for health-related selection.

In addition to these two directions of causality, there is also the possibility of third variables (such as cognitive ability or personality traits) influencing both. Some studies have indeed shown that differences in cognitive ability and other personal attributes—whose formation plausibly predates the attainment of various income levels in adult life—explain part of the income-related inequalities in health. There is also emerging evidence of genetic determinants of income and material deprivation, again probably acting through cognitive ability and other personal attributes.³¹⁶

Quasi-experimental evidence

For an assessment of a causal effect of income on health it is therefore best to rely on experimental and quasi-experimental studies, but it is important to recognize from the outset that most of these studies have a number of important limitations. They often study the effect of rather small variations in income, sometimes in settings (such as lotteries or stock market gains) that may not represent the experience of a lower or higher regular income over longer periods of life. It is also more difficult to demonstrate causation (i.e. the effects of a change in income on health, which may take a long time to materialize) than to demonstrate reverse causation (i.e. the effects of a 'health shock' on income, which can be seen within a couple of years).

Nevertheless, experimental and quasi-experimental studies potentially provide the strongest evidence for a causal effect of income on health. Recently, a number of reviews on these types of studies have been carried out. Overall, the main conclusion is that in high-income countries there is clear evidence for a causal effect of major changes in health on income, but there is no consistent evidence for a causal effect of modest and short-term changes in income on physical health in adulthood. However, all reviews emphasize that the available evidence does not rule out the possibility that there is a causal effect of larger variations in lifetime income on physical health in adulthood. Also, they agree that the evidence for a causal effect of parental income on the health of children is more consistent^{238,295,314,317,318}

The evidence for reverse causation is generally considered convincing. In a range of studies exploiting 'exogenous' changes in health (i.e. health events that are abrupt and unforeseen), ill-health in adulthood had a modest negative effect on wages among those who work, and a stronger effect on income through decreasing the employment rate and reducing the hours worked among the employed.³¹⁴

In addition, ill-health in early life and childhood had substantial effects on lifetime earnings, through decreasing the build-up of cognitive and non-cognitive abilities, constraining the acquisition of education, and by continuing into ill-health in adulthood which then interferes with labour productivity in adulthood. Ill-health can thus have a very long reach from childhood to constrained economic opportunities in adulthood.³¹⁴

On the other hand, studies trying to find evidence for a causal effect of income on physical health in adulthood in high income countries using a quasi-experimental set-up have had inconsistent

results. One review summarizing the results of 16 studies found eight studies with no effect, two studies with a negative effect (i.e. more money, worse health), and six studies with a positive effect (i.e. more money, better health). Based on a further evaluation of the methodological quality of these studies, the authors conclude that the evidence that income does have a causal impact on health in adulthood is ‘weak’.³¹⁴ For low- and middle-income countries the evidence for a causal effect of income on health—which partly comes from true experiments—was considered to be more convincing.³¹⁴

A second recent review summarizing the results of nine studies of income effects on health in adulthood (six of whom were also included in the first review) found four studies with no effect, two studies with a negative effect, and three studies with a positive effect. When looking at other outcomes, the review did find strong evidence that additional financial resources during adulthood make people happier and reduce mental health problems, but also that more money can lead to less healthy behaviours such as more drinking and smoking. The authors conclude that for physical health in adulthood the evidence is ‘mixed’.³¹⁸

Some of the reviewed studies focused on so-called windfall gains in income, for example, lottery winnings, which closely approximate a true experimental setting. Some European studies found that the recipients of lottery prizes experience positive changes in self-reported health. These positive effects are particularly seen for mental health and less so for physical health, perhaps because winning a lottery also tends to increase smoking and drinking.³¹⁴

The weakness of this research strategy is, of course, that this variation in ‘income’ does not necessarily correspond to that of normal monthly or yearly income. This limitation has to some extent been circumvented in a recent study of a Swedish lottery, not included in the reviews quoted above, that distributed sizable prizes and paid them out over longer periods of time, but this study also found largely null effects on physical health in adulthood.³¹⁹

The reviews also agree that the evidence for a causal effect of parental income on children’s health is considerably stronger than that for adults’ income on their own health. As the likelihood of reverse causation is less, because children’s health will not directly affect their parents’ income, observational evidence does not have to be discarded altogether, provided there is sufficient control for confounding by third variables. Although evidence from experimental and quasi-experimental studies is again somewhat mixed, reviews conclude that a causal effect of parental income on children’s health is likely to exist.^{295,314,317}

This conclusion is further supported by the fact that parental income has positive effects on intermediate outcomes, such as the quality of parenting, the physical home environment, maternal depression, smoking during pregnancy, and children’s cognitive ability, school achievement, and behaviour.³¹⁷ Long-term increases in incomes of lower socioeconomic groups may thus have health benefits that accumulate over generations.²³⁸

Nevertheless, the lack of convincing evidence for an effect of income on adult physical health in high-income countries makes one wonder whether this does not partly resolve the paradox of the persistence of health inequalities in modern welfare states. Has the welfare state perhaps reduced the importance of one mechanism—low income causing ill-health—so that we are now mainly left with other mechanisms—ill-health leading to low income, and confounding by cognitive ability and other personal characteristics?

Provisional conclusions

Surprisingly, after decades of research, there is still uncertainty about whether there is a causal effect of socioeconomic position on health, and if so about how much of the observed association between socioeconomic position and health is due to such a causal effect.

Social epidemiologists have generally tended to interpret the association between socioeconomic position and health as being largely due to a causal effect of socioeconomic position on health. However, recent research in other disciplines, such as genetics and economics, has cast doubt on this interpretation. The new evidence reviewed above suggests that both health-related selection (in the case of the association between income and health) and confounding by unobserved personal attributes (in the case of the association between education, occupational class, and income) play a more important role than previously recognized.

At the same time, the importance of the new (quasi-)experimental evidence should not be overrated. As mentioned above, the paradox here is that the closer we get to identifying a true causal effect, the farther away we may get from a good understanding of how socioeconomic position affects health. We have to sail between the Scylla of being strict on causality (but not capturing the full effects of living in socioeconomic disadvantage) and the Charybdis of more fully capturing the effects of living in socioeconomic disadvantage (but being too lenient on causality). In other words, **the new evidence may well underestimate the full causal effects of a life spent in socioeconomic disadvantage.**

Nevertheless, at this stage of advancing knowledge we cannot claim to know that all or even most of the associations between socioeconomic position and health are due to a causal effect of socioeconomic position on health. **The most we can say is that there is a causal effect of education on health, and that the observed association between education and health is therefore partly due to such a causal effect. It is also likely that there is a causal effect of parental income on children's health, and while there is no clear evidence for an effect of modest changes in income on adult health, it is plausible that larger changes in income, particularly at the low end of the income distribution and particularly in countries with lower average incomes, do have an effect on adult health.**

3.3 Six groups of contributing factors

Life-course models

Six groups of specific factors are likely to play a role in the explanation of health inequalities—although not necessarily in a strictly ‘mediating’ role (see section 3.1 ‘Methodical issues’). These are: genetics, childhood environment, material living conditions, social and psychological factors, health-related behaviours, and health care.

Before we discuss each of these in turn, we introduce the life-course perspective that will be used to structure the presentation—and after presenting the contributing factors we end by briefly discussing the biological pathways linking socioeconomic position to health. This has resulted in this section being very long, reflecting the enormous productivity of research in this area in the past three decades.

What transpires from all this research is a rather complex picture of how individuals in lower socioeconomic groups are exposed over their lifetime to a wide variety of unfavourable living conditions, and how these exposures and the behavioural responses to them lead to ill-health—but also of how health influences socioeconomic position and how both are influenced by personal characteristics, like cognitive ability and personality traits, which partly reflect childhood environment and are also partly genetically determined. All these bits and pieces can be brought together in a ‘life-course’ perspective.

At conception, all individuals are endowed with a particular set of genes, and after birth move from a situation that is still largely determined by their parents’ socioeconomic position to one that is determined by their own socioeconomic position. They enter the labour market with

certain educational credentials, and during their adult life move through various occupations and varying levels of income, reaching retirement with or without some wealth. During each of these life stages health problems may be a consequence of their previous and current socioeconomic position, and a determinant of their current and future socioeconomic position. Moreover, both health and socioeconomic position may be determined by personal characteristics that are themselves the consequences of socioeconomic conditions in previous stages of life.

A life-course approach then sees the higher rates of illness and premature death among adults and older persons in lower socioeconomic groups as the cumulative result of socially patterned exposures acting at different stages of the life-course, and at the same time as possible determinants of future changes in socioeconomic position.³²⁰ This has proven to be a very useful way of integrating different strands of evidence.

The simplest conceptual model for life-course influences is that of ‘accumulation of risk’. Different forms of material and immaterial disadvantage tend to cluster in the same persons, with one disadvantage increasing the likelihood of another at a later point in time, and with health disadvantage arising as a result of cumulative social disadvantage.^{321,322} Such accumulation models can incorporate both ‘selection’ and ‘causation’ mechanisms, because a low socioeconomic position in one stage of the life-course may translate into a health disadvantage in the next, which may then lead to a still lower socioeconomic position some years later, and so on.³²³

Life-course models may also incorporate ‘critical periods’: time windows of exposure that are particularly important for health at later ages. One possible example of a ‘critical period’ is intra-uterine life, as elaborated in the ‘foetal origins of adult disease’ hypothesis which holds that foetal growth and other factors during pregnancy are linked to the risk of developing cardiovascular and other diseases in adulthood.³²⁴ Another is childhood: the child’s physical, cognitive, and emotional development is strongly influenced by socioeconomic circumstances, which in its turn influences both adult socioeconomic position and adult health in many ways (‘the long arm of childhood’).³²⁵

Circumstances in early life also set up a pattern of social learning, which may generate a sense of powerlessness reinforced by others in the social network who have been similarly disadvantaged and socially excluded, sometimes over generations.³²⁶ Such intergenerational transmission of social and health disadvantage may therefore be one of the mechanisms contributing to the persistence of health inequalities over time.

Genetics

A life-course approach to health inequalities starts by clarifying the role of genetics.^{ix} Investigating the role of genetics in generating social and health inequalities is not only scientifically challenging—Box 3.2 listed some of the complexities—but also surrounded by intense dispute, fuelled by fears that research findings may be (mis)used to justify existing inequalities.³²⁷ However, as we will see in Chapter 5, inequalities originating in genetic differences are not necessarily less inequitable than inequalities based on differences in living conditions, and even if they were we cannot close our eyes to the results of genetic studies.

These results clearly show that genetic factors are likely to play a role in generating health inequalities, by partly determining cognitive ability and other personal characteristics, which in their turn also partly determine both a person’s socioeconomic position and his or her health. Technically speaking, genetically determined variations in these personal characteristics are confounders of the relationship between socioeconomic position and health, strengthening a relationship that would otherwise have been present in a weaker form.

The important role of genetic determinants for children's cognitive ability has first been convincingly shown in twin studies, which found substantial heritability in the order of at least 50%.^{278,328} Molecular studies have more recently identified many specific genetic variants that are associated with cognitive ability,^{278, 279} and although there is still a big gap between the combined effects of all genetic variants that have until now been identified, and the heritability estimated from twin studies, this 'heritability gap' is being slowly filled as the findings of more molecular studies accumulate.^{277,280}

The transmission of genetic material from parent to child should not be seen as a completely pre-determined process. Cognitive ability and personality are complex traits that are determined by many genes, and a complete set of genes associated with a particular level of intelligence or personality profile cannot be transmitted intact from parent to child, because the relevant genes are located on different chromosomes and will become re-assorted during cell division and meiosis.³²⁹ Nevertheless, as a result of both intergenerational transmission and random re-assortment children do differ in their genetic predisposition towards low or high cognitive ability and towards various personality profiles.

The role of genetics in the explanation of health inequalities is not limited to 'gene-environment correlation' (i.e. genetic factors occurring more frequently in lower than in higher socioeconomic groups or vice versa), but may also include 'gene-environment interaction' (i.e. genetic factors determining the effects of the socioeconomic environment).^{330,331} Some genotypes may increase or decrease the susceptibility to a disadvantaged environment,³³² and in twin studies the heritability of cognitive ability has been found to be larger in higher than in lower socioeconomic groups, probably because the effect of the environment overwhelms the effect of genetic determinants in lower socioeconomic groups.³³³

The role of genetics in explaining health inequalities may also involve 'epigenetics', that is, heritable changes in gene function that do not involve changes in the DNA sequence, for example, due to methylation of DNA. Such changes may be the result of various exposures, such as smoking, nutrition, psychosocial stress, and environmental toxicants, and may play a role in the generation of health inequalities and their intergenerational transmission.^{334,335} However, epigenetic mechanisms would be an instance of mediation, not of confounding.

Childhood environment

Moving on through the life-course, what happens after conception is pregnancy and childhood. Whether or not pregnancy and childhood conditions can mediate the effect of socioeconomic position on health, depends on whether we look at health inequalities in childhood or in adulthood. In the latter case—which is what we are focusing on in this chapter—they logically cannot, because pregnancy and childhood conditions temporally precede adulthood. But if we take into account the fact that a person's socioeconomic position often has a high degree of continuity over the life-course, it does certainly make sense to consider the role of pregnancy and childhood conditions as contributing factors.

This continuity is illustrated in Figure 3.1—which, however, also shows that the degree of continuity of socioeconomic position over the life-course should not be exaggerated. On the one hand, many more low than high educated people come from lower occupational class families. On average in Europe as a whole, 58% of low educated men had a father with a manual occupation, against 35% of high educated men. This continuity indicates that inequalities in exposure to socioeconomic disadvantage during childhood may indeed play a role in explaining health inequalities in adulthood.

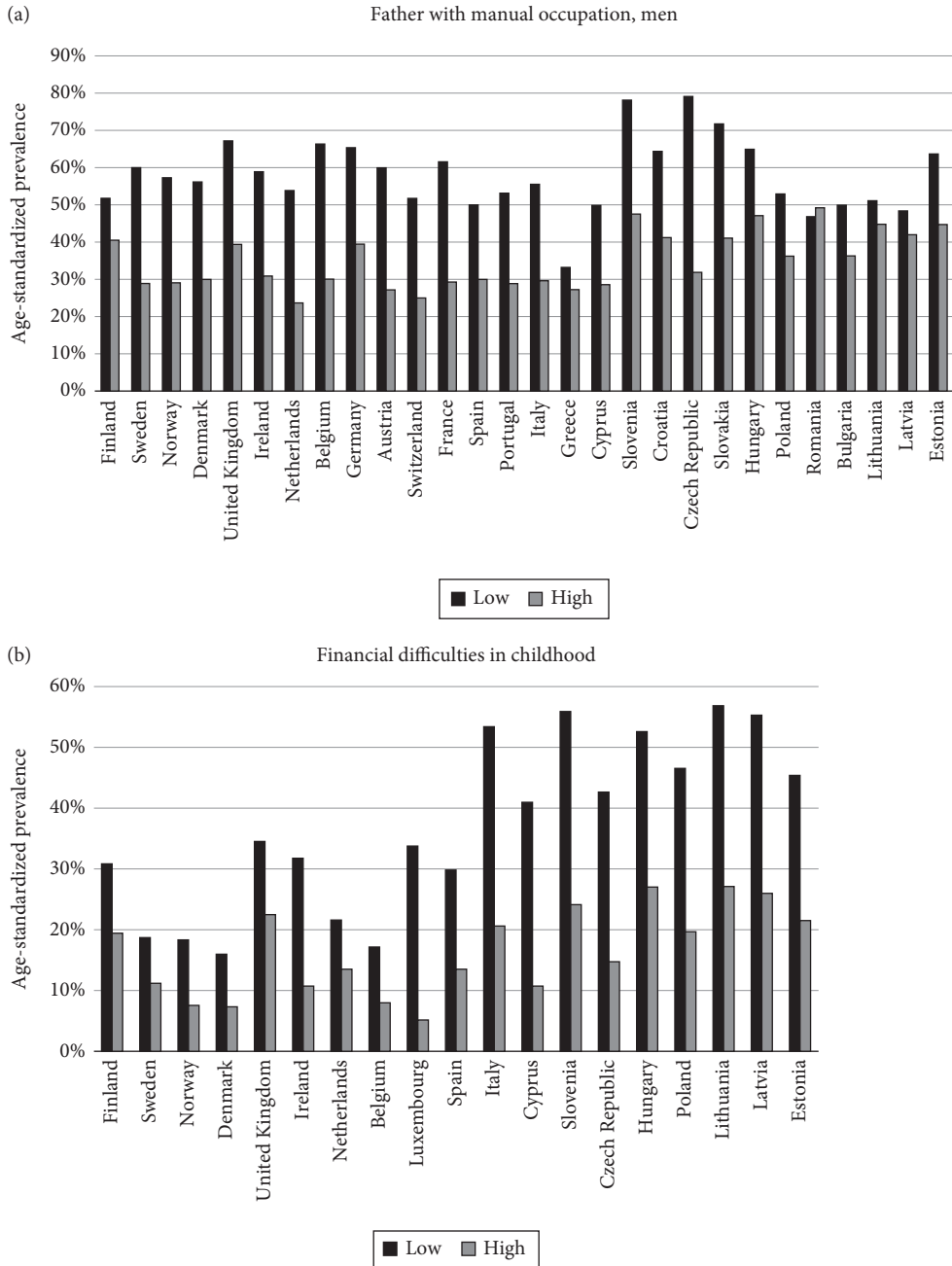


Figure 3.1 Educational inequalities in childhood conditions

Notes for panel a: Men aged 35–79 years (women similar). Data for ca. 2010.

Source: dataset constructed in DEMETRIQ/LIFEPATH projects, with harmonized data from European Social Survey.

Notes for panel b: Financial hardship = self-reported financial difficulties (most of the time/often) when growing up. Women aged 35–79 years (men similar). Data for 2005.

Source: dataset constructed in DEMETRIQ/LIFEPATH projects, with harmonized data from EU Statistics on Income and Living Conditions.

On the other hand, not all low educated men had a father with a manual occupation, and only 31% of high educated men had a father with an upper non-manual occupation—which really counts as growing up in advantage. Thus, although there is a certain degree of continuity in socioeconomic position, this is diluted by social mobility.

Adverse childhood conditions

Socioeconomic conditions already have an impact on children's health and development before they are born. Low socioeconomic position of the mother is associated with impaired foetal growth, low birth weight, and premature birth,^{336,337} which may all contribute to a higher frequency of health and developmental problems after birth—and perhaps even into adulthood.³²⁴ Mechanisms linking socioeconomic disadvantage of the parents to impaired foetal growth include impaired maternal health (e.g. gestational hypertension and diabetes), more psychosocial stress (e.g. due to financial and relational problems), and parental health behaviours (e.g. smoking during pregnancy).³³⁸

After birth, there are also substantial inequalities in exposure to a range of unfavourable childhood conditions. Retrospectively, having experienced serious conflict or hardship in childhood is much more common among adults with a lower than with a higher socioeconomic position. This is illustrated in panel B of Figure 3.1, which also shows that financial hardship in childhood is reported much more frequently in Southern, Central-Eastern and Eastern Europe, probably reflecting lower levels of prosperity in the past.

Systematic reviews show that growing up in disadvantaged socioeconomic circumstances is associated with many negative health and development outcomes in childhood, ranging from impaired growth to general ill-health and from impaired neurocognitive functioning, structural brain development, and mental health to higher rates of asthma and dental caries.^{107,283,339–341}

This is commonly thought to reflect a causal effect of socioeconomic disadvantage on health, because there is less potential for reverse causality in the case of children's health and development than in the case of adult health outcomes.⁴⁷ However, it is also possible that there are common underlying factors in the association between parents' socioeconomic disadvantage and children's health and development, such as genetic factors shared by parents and their children.²⁸⁵

Systematic reviews also show that growing up in disadvantaged socioeconomic circumstances is associated with many negative health outcomes in adult life, independent from the socioeconomic position children achieve for themselves, including higher increased all-cause mortality, mortality from various specific causes, cardiovascular risk factors, impaired cognitive and physical functioning, and less good self-rated health.^{342,343}

Several mechanisms are likely to be involved in these delayed effects.¹⁰⁷ Health inequalities initiated in childhood may continue into health inequalities later in life when ill-health in childhood continues into ill-health in adulthood, and even more so when it also negatively affects adult education or income.³⁴⁰

Adverse childhood conditions may also contribute to health inequalities at older ages through other, more indirect mechanisms.^{344,345} Growing up in unfavourable socioeconomic conditions impairs the neurocognitive development of children, particularly their language ability and executive functioning (i.e. the cognitive processes necessary for the control of behaviour, such as attentional control and working memory).^{341,346} Neuroimaging studies have even found a biological substrate for some of these effects, such as reduced grey matter volumes of certain parts of the brain.³⁴⁷

These effects of socioeconomic disadvantage on neurocognitive development are probably due to a combination of prenatal exposures (such as higher levels of maternal stress hormones), parental care (such as harsh discipline), and lack of cognitive stimulation.³⁴⁸ Stimulation of learning

by parents raises children's intelligence, and the comparatively harsh living conditions of families with a lower socioeconomic position increase family stress and hamper family investments in children, which harm the development of their cognitive ability and personality.³⁴⁹

The socioeconomic environment in which children grow up may also have an impact on their epigenome with effects on health and development throughout the life-course.³⁵⁰ For example, some studies have found an association between childhood socioeconomic status and DNA methylation of genes regulating stress reactivity and inflammation, suggesting a specific biological pathway linking childhood disadvantage to adult ill-health.^{351,352}

Simpler mechanisms may, however, also play a role. For example, the higher prevalence of smoking in lower socioeconomic groups can partly be explained by the fact that children from low educated or poor parents copy the behaviour of their parents, or do not acquire the skills or motivation to resist the temptations of smoking.³⁵³

Cognitive ability and personality

At this juncture, after having presented some evidence on the roles of both genetics and childhood environment, we can also more fully discuss the role of personal characteristics such as cognitive ability and personality. As a result of both genetics and childhood environment, adults in lower and higher socioeconomic groups differ in a number of personal characteristics that promote good or bad health, and that contribute to health inequalities. This has been convincingly documented for both cognitive ability and personality traits.

'Cognitive ability' or 'intelligence' has been defined as 'a very general mental capability that involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience.'³⁵⁴ Although cognitive ability has many dimensions, as illustrated by this definition, these are usually strongly inter-correlated, which has given rise to the idea that the underlying general ability (often called 'g') can be measured along a single scale of intelligence.

On average, people in lower socioeconomic groups have substantially lower scores on intelligence tests. This is not only true when we group people by their level of education, but also when we group them by their occupational class or level of income.³⁵⁵ For example, in Britain the average scores on a general intelligence test were found to be almost twice as high for men in professional occupations than for men in unskilled manual occupations,³⁵⁶ and in Sweden one third of men in unskilled manual occupations was found to have a very low intelligence test score against only one tenth of men in the whole population.³⁵⁷

As explained above, differences in cognitive ability reflect a mixture of the effects of genetics and childhood environment, but regardless of the ultimate causes children with low cognitive abilities are less upwardly mobile,^{358,359} and thus are over-represented in lower socioeconomic groups as adults.

Lower cognitive ability is also very strongly and independently associated with a range of unfavourable health outcomes, from health-related behaviours and specific diseases and injuries to all-cause mortality.^{360 361, 362} A recent meta-analysis estimated that one standard deviation higher cognitive test score in childhood or youth was associated with 24% lower mortality during several decades of follow-up.³⁶³

As was to be expected in view of these strong interrelations, differences in cognitive ability have been found to account for a sizable part of socioeconomic inequalities in health.^{357,362,364} For example, a long-term follow-up study of schoolchildren in the Netherlands who had undergone intelligence testing in the 1940s, found that about half of the observed inequalities in mortality by level of attained education could be explained by differences in cognitive ability as measured in the last grade of primary school.³⁶⁵ In the British Whitehall study about one third of inequalities

in various health outcomes by 'employment grade' (a measure of occupational class among civil servants) could statistically be explained by differences in cognitive ability.³⁶²

People in lower and higher socioeconomic groups also differ in their personality profiles as measured with the so-called Big Five personality traits. People in lower socioeconomic groups have been found to have higher levels of 'neuroticism' and 'agreeableness', and lower levels of 'extraversion', 'openness', and 'conscientiousness'. Because some of these factors are also predictors of mortality, differences in personality statistically explain part of inequalities in mortality.^{366,367}

Material living conditions

Poverty

Although differences in income level across the whole income distribution may not play an important role in generating health inequalities in high income countries (as discussed in section 3.2 'Education, occupation, income, and health'), poverty does plausibly play a role, but to a different extent in different European countries, depending on its depth and impact on access to other resources important for health.

Poverty can be measured in different ways, for example, as a net household income level below 60% of the median, or below a threshold deemed to represent the minimum income necessary for a decent life.³⁶⁸ It can also be measured more directly as material deprivation, in the form of lack of access to a number of items deemed necessary for a decent life (e.g. having a telephone, having a refrigerator, being able to receive friends, having at least one week of holiday per year, etc.).

Figure 3.2 shows that the prevalence of material deprivation is very unevenly distributed between people with lower and higher socioeconomic positions. Although there is some material deprivation among the high educated, its prevalence is much higher among the low educated. The gap in material deprivation between low and high educated differs substantially between countries: countries with more generous welfare policies, as indicated by higher social expenditure (see Figure 1.2) tend to have lower levels and smaller inequalities in material deprivation.

Many studies show that poverty is associated with a range of adverse health outcomes,^{369,370} and a few mediation analyses which link indicators of socioeconomic position to health outcomes via poverty suggest that differences between socioeconomic groups in the prevalence of poverty do indeed contribute to the explanation of health inequalities.^{371–373}

The plausibility of a causal effect of poverty on health is supported by the existence of a range of well-documented pathways through which poverty may affect health. It reduces financial access to activities and products that are important for the maintenance and promotion of health (such as a healthy diet, sports, social contacts). This includes reduced access to health care services, particularly when out-of-pocket payments are required. And it often leads to psychosocial stress, which has negative biological and mental effects and increases the likelihood of risk-taking behaviours (such as smoking and excessive alcohol consumption).^{368,374,375}

Work and working conditions

A second group of living conditions contributing to health inequalities are working conditions. Poor working conditions are more prevalent among employed people with lower levels of occupation: many studies have shown physical, chemical, and psychosocial exposures to be more common in lower occupational groups.^{376,377} Figure 3.3 illustrates this for one harmful physical working condition (carrying heavy loads) and one unfavourable psychosocial working condition (not being able to choose or change the order of one's tasks). Both are much more frequent in the lower occupational classes in all European countries.

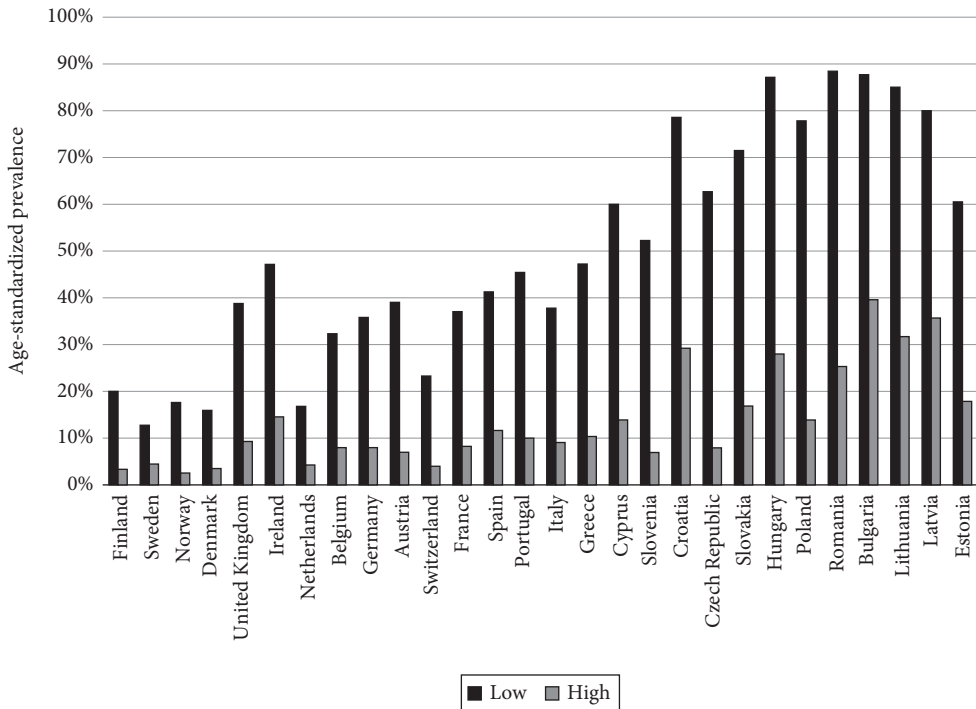


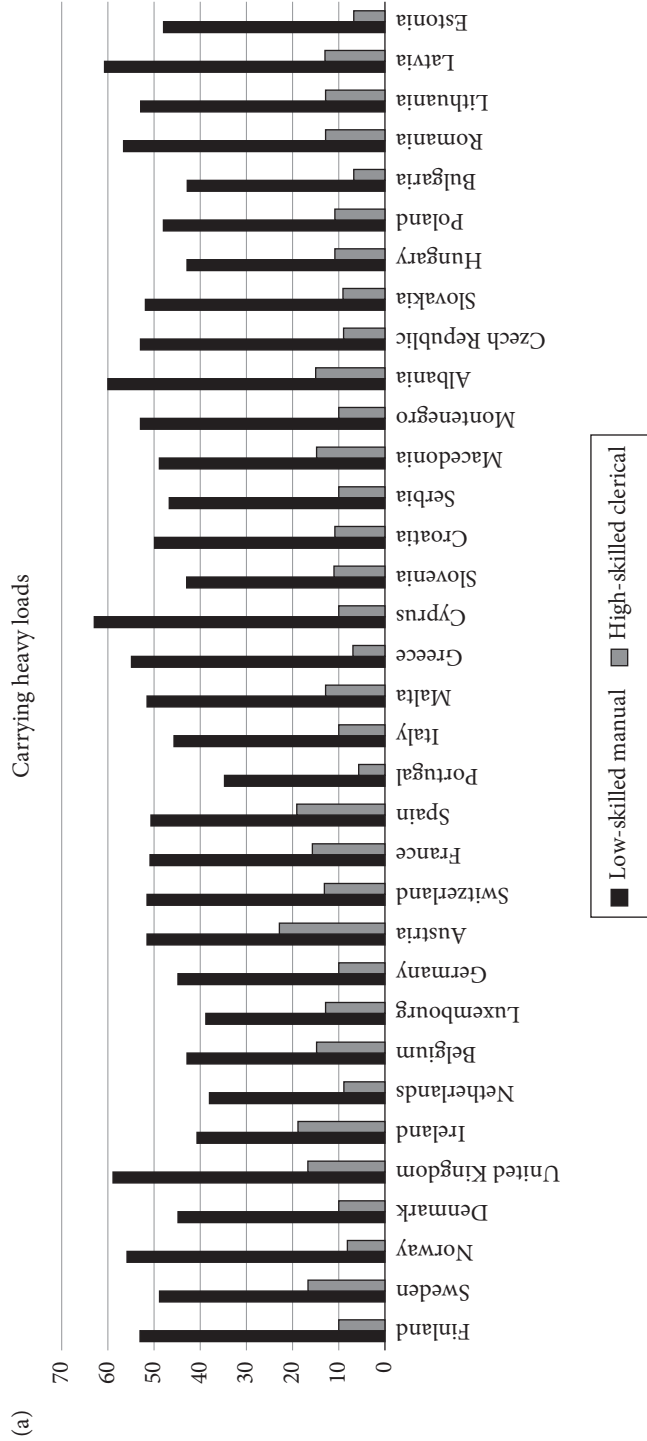
Figure 3.2 Educational inequalities in material deprivation

Notes: Material deprivation = at least 3 out of 9 necessities missing. Men only; women similar. Data for 2010.

Source: dataset constructed in DEMETRIQ/LIFEPATH projects, with harmonized data from EU Statistics on Income and Living Conditions.

There is also moderate to good evidence that these exposures lead to various forms of ill-health, and a recent systematic review of mediation analyses using the 'difference method' confirms a role in generating health inequalities for both the physical/chemical work environment (higher exposure to physical demands, biomechanical strains, and chemical substances in certain lower occupations) and the psychosocial work environment (higher prevalence of demand-control and effort-reward imbalance in certain lower occupations).³⁷⁸ An imbalance between effort and reward at work has also been found to have a mediating role in the higher prevalence of depressive symptoms in lower occupational classes.³⁷⁸

Under the broad heading of working conditions we also need to consider unemployment and precarious employment. These conditions are more common in lower socioeconomic groups.³⁷⁹ The health effects of unemployment have been a long-standing issue for scientific debate, partly because there is strong selection of unhealthy persons out of employment. Such selection is, however, less pronounced during mass unemployment, and several studies have used this insight to look at health consequences during and after episodes of mass unemployment. It appears that deep recessions do cause excess mortality among those who experience unemployment, for instance from suicide, alcohol-related conditions, and cardiovascular diseases, in particular if the unemployed have a low education.^{380–382} Moreover, precarious employment, high job insecurity, and wages that are low or perceived to be unfairly paid are associated with elevated risks of stress-related disorders.³⁷⁹



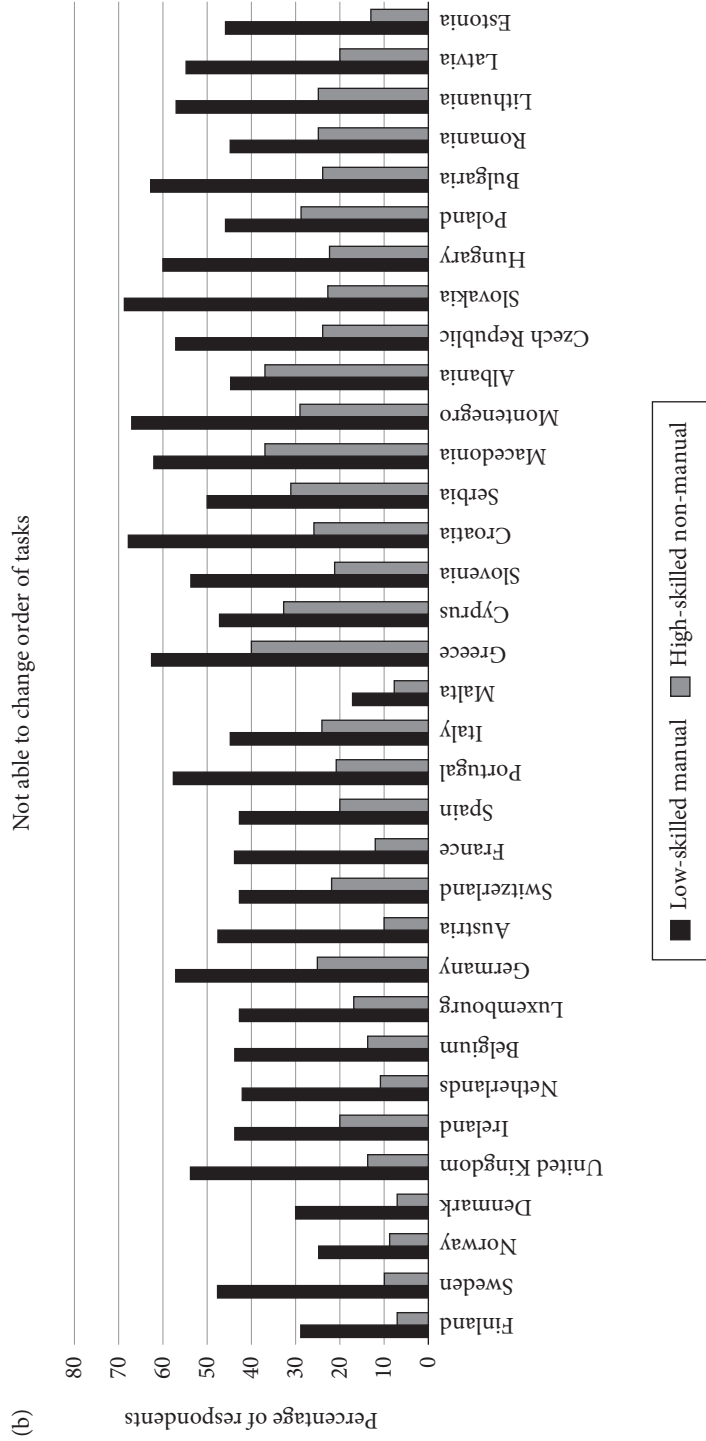


Figure 3.3 Occupational class inequalities in working conditions

Notes: Carrying heavy loads = Q. 'Does your work involve carrying or moving heavy loads?' A: 'Yes, at least ¼ of the time.' Not able to change order of tasks = Q: 'Are you able to choose or change your order of tasks?' A: 'No.' Only two occupational classes shown. Men and women combined.

Source: data from European Working Conditions Survey (<https://www.eurofound.europa.eu/data/european-working-conditions-survey>, accessed July 2018).

Housing, neighbourhood, wider environment

A third group of factors are the conditions under which people live, both in the immediate sense of **quality of housing** (e.g. crowding, dampness, and accident risks),³⁸³ and in the more indirect sense of their **neighbourhood conditions** (access to amenities such as sports facilities and groceries with fresh fruit and vegetables, presence of psychosocial stressors, levels of social capital),^{384–386} all of which tend to be less favourable for lower socioeconomic groups. People with a **low socioeconomic position also tend to be more exposed to general environmental health risks such as air pollution and toxic waste dumps.**^{387,388}

Social and psychological factors

Another group of specific determinants which likely contribute to the explanation of health inequalities are non-material factors such as social and psychological factors.

Psychosocial stress

People with a **low socioeconomic position on average are exposed to more psychosocial stressors than people with a high socioeconomic position, in the form of more negative life events** (e.g. loss of beloved ones or loss of paid work), **more ‘daily hassles’** (e.g. in the form of financial difficulties), **and a combination of high demands and low control in life as a whole.**³⁸⁹ At the same time, they also **tend to have smaller social networks, lower levels of social support, and less ‘social capital’ generally,**³⁹⁰ as well as **less effective coping styles** (e.g. a more external ‘locus of control’).³⁹¹ **People with a higher socioeconomic position have a stronger sense of control over their own lives, including their work, and this is associated with healthier behaviour, and lower rates of morbidity and mortality.**^{392,393}

This combination of a higher exposure to psychosocial stressors and less capacity to remove or buffer these exposures may explain part of the higher frequency of health problems in the lower socioeconomic groups. At least two pathways may be involved. The first is a behavioural pathway: **psychosocial stress and other unfavourable psychosocial factors increase the likelihood of unhealthy behaviours, such as smoking, excessive alcohol consumption and lack of physical exercise.**^{394–397}

The second is a more direct biological pathway. The experience of **stress affects the neural, endocrine, and immune systems of the body, and chronic stress may lead to maladaptive responses in the form of, for example, high blood pressure, a prolonged high level of cortisol, higher blood viscosity, or a suppression of the immune response,** which may in turn increase the susceptibility to a range of diseases.^{398–400} We will further discuss the biological pathways linking low socioeconomic position to ill-health at the end of this paragraph.

Although the evidence suggests that higher exposure to psychosocial stressors, in combination with less capacity to remove or buffer these exposures, has negative health effects, consensus has not yet emerged.⁴⁰¹ If there is an independent health effect of psychosocial factors—which seems plausible—then this may explain part of the higher frequency of health problems in lower socioeconomic groups. This has been best documented for psychosocial factors related to work organization, such as job strain, which have been shown to contribute to socioeconomic inequalities in cardiovascular health, including in mediation analyses.³⁷⁸ **Mediation analyses focusing on the role of psychosocial stressors outside the work environment are less common, but have also suggested a role for psychosocial factors in generating health inequalities.**^{371,402}

Other non-material factors

The role of non-material factors is not limited to psychosocial stressors and their moderators. Socioeconomic groups differ profoundly in their exposure to a wide range of social, cultural, and

psychological factors, use of media like television and the internet, living with a partner, having someone to confide in, loneliness, trust, being religious, feeling depressed, life satisfaction, happiness, etc.⁴⁰³

Figure 3.4 shows this for a few examples. The low educated are almost always worse off, but the magnitude of these inequalities differs importantly between European regions. Northern Europe almost always has a lower average prevalence of unfavourable conditions, but relative and/or absolute inequalities are not always smaller than in other European regions, suggesting that the welfare state has probably reduced material inequalities more than non-material inequalities. ●

Health-related behaviours

The role of health-related behaviours, such as smoking, excessive alcohol consumption, inadequate diet, lack of physical exercise and obesity, in generating health inequalities has been relatively well documented. These are established causal determinants of morbidity and mortality, and are often more prevalent in the lower socioeconomic groups in many high-income countries.^{132,404–407} This is also the group of factors for which most formal mediation analyses have been carried out, which generally show that health-related behaviours make substantial contributions to the explanation of health inequalities.⁴⁰⁸ Despite the fact that almost all mediation analyses followed the conventional ‘difference method’, so that we cannot attach too much importance to their quantitative results, a non-trivial role of health-related behaviours in generating health inequalities is highly plausible.

Smoking

By far the most widely available data on a specific determinant of health inequalities relate to smoking. In many European countries cigarette smoking is the leading determinant of health problems. Systematic reviews have shown the prevalence of smoking to differ strongly between socioeconomic groups in many high-income countries,^{404,409} particularly among men, and mediation analyses have found that smoking alone accounts for a substantial part of socioeconomic inequalities in mortality.^{408,410,411}

There are, however, major differences between European countries in the magnitude of inequalities in smoking, and consequently in the contribution of smoking to inequalities in mortality and other health outcomes.^{262,412} As has been shown in a number of comparative studies, inequalities in smoking follow a North-South gradient within Europe, with larger inequalities in the North and smaller (sometimes even ‘reverse’) gradients in the South.^{413,414}

This is particularly clear in the case of women: high educated women smoke considerably less than low educated women in the North of Europe, but the reverse is true in several Southern European countries, such as Portugal, Italy, and Cyprus (Figure 3.5).

Among men, the gap between low and high educated is large in the North and West of Europe, not so much because the prevalence of smoking among the low educated is so high, but primarily because the prevalence among the high educated is so low. The gap is smaller in Spain, Portugal, and Italy, mainly because the prevalence among the high educated is so high. Among women, the gap is small in Southern Europe too, partly because smoking prevalence is low among the low educated, and partly because it is high among the high educated (Figure 3.5).

This all fits with the idea that Southern European countries are in an earlier phase of the ‘smoking epidemic’, in which smoking among high educated men and women has not yet declined strongly, and in which smoking among low educated women has not yet risen strongly.⁴⁰⁹ The concept of the smoking epidemic will be further discussed in section 3.4 ‘Theories about the explanation of health inequalities’.

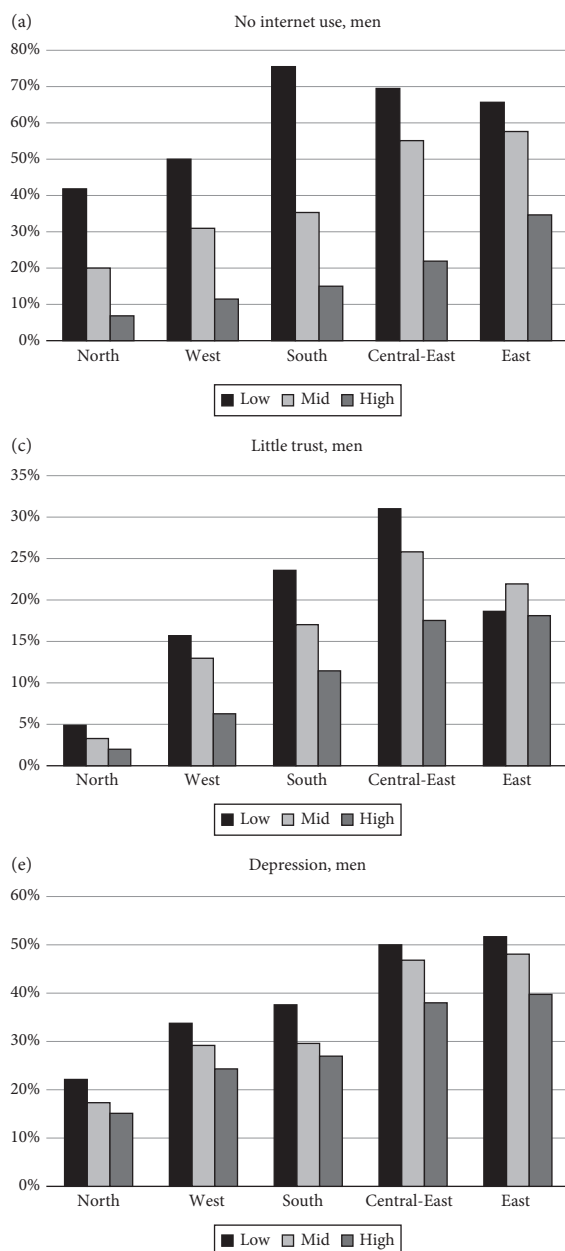


Figure 3.4 Educational inequalities in social, cultural and psychological factors

Notes: The survey questions are given below:

No internet use = Q: 'Personal use of internet/e-mail/www'; A: 'No access' or 'never use'. Not living with partner = Q: 'Lives with husband/wife/partner in same household'; A: No. Little trust = Q: 'Most people can be trusted or you can't be too careful'. A: 0,1,2 on scale of 11. Security values = Positive score on scale consisting of two items: 'It is important to him to live in secure surroundings. He avoids anything that might endanger his safety' and 'It is very important to him that his country be safe from threats from within and without. He is concerned that social order be protected.'. Depression = Q: 'Felt depressed, how often last week'; A: 'Some of the time', 'Most of the time' and 'All or almost all of the time'. 'Little life satisfaction' = Q: 'How satisfied with life as a whole'; A: 0–4 on a scale from 0–10.

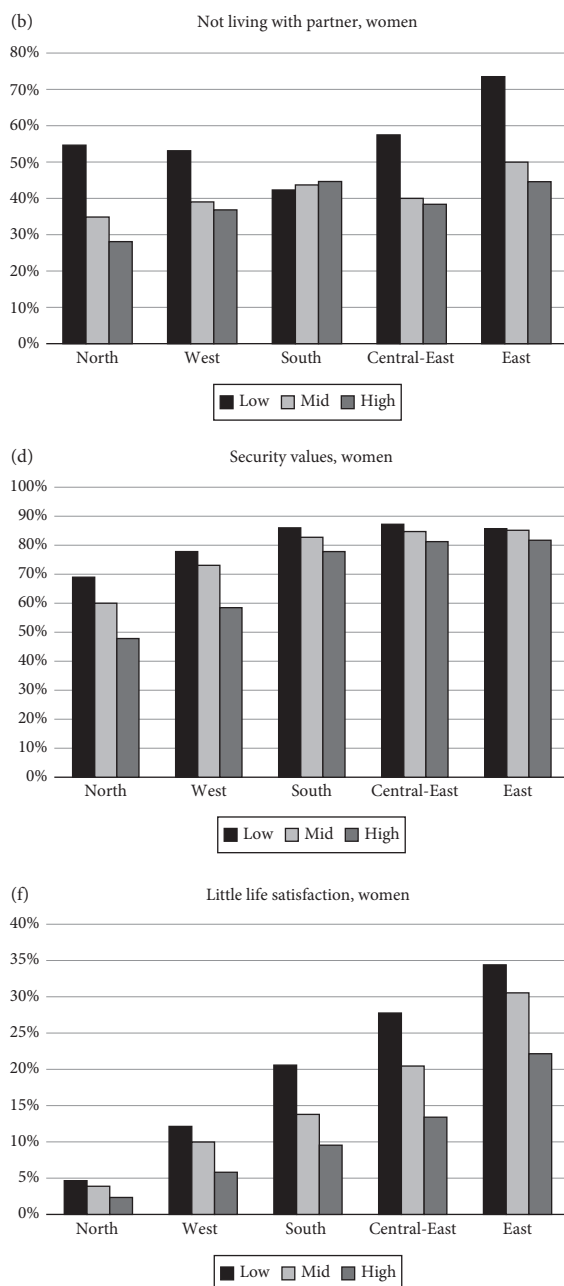
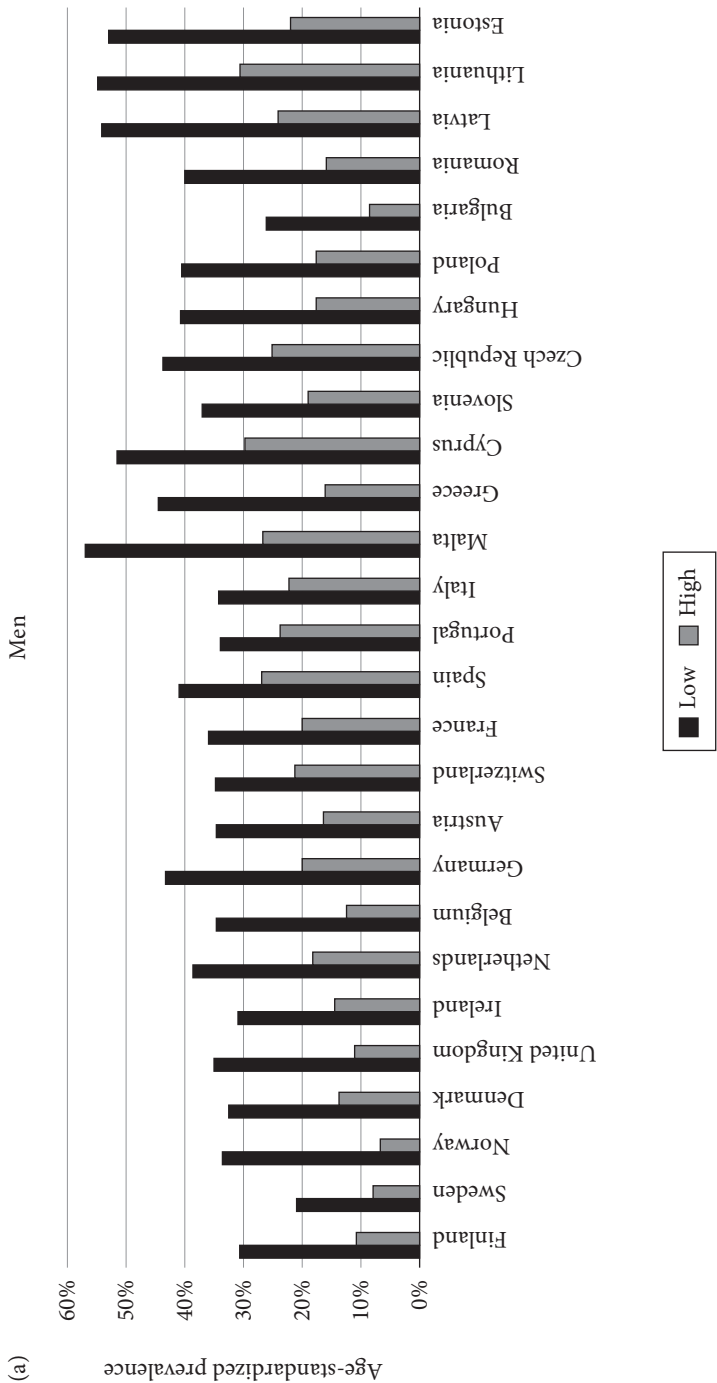


Figure 3.4 Continued

Men and women show similar findings for all variables. Pooled dataset of 25 countries. North = Finland, Sweden, Norway, Iceland, Denmark. West = Austria, Belgium, France, Germany, Ireland, Netherlands, Switzerland, United Kingdom. South = Portugal, Spain, Italy, Greece, Cyprus. Central-East = Croatia, Czech Republic, Hungary, Poland, Slovakia, Slovenia. East = Estonia, Lithuania, Russia, Ukraine.

Source: data from European Social Survey cumulative data file 2002–14 (<https://www.europeansocialsurvey.org/data/>, accessed 2 April 2017)



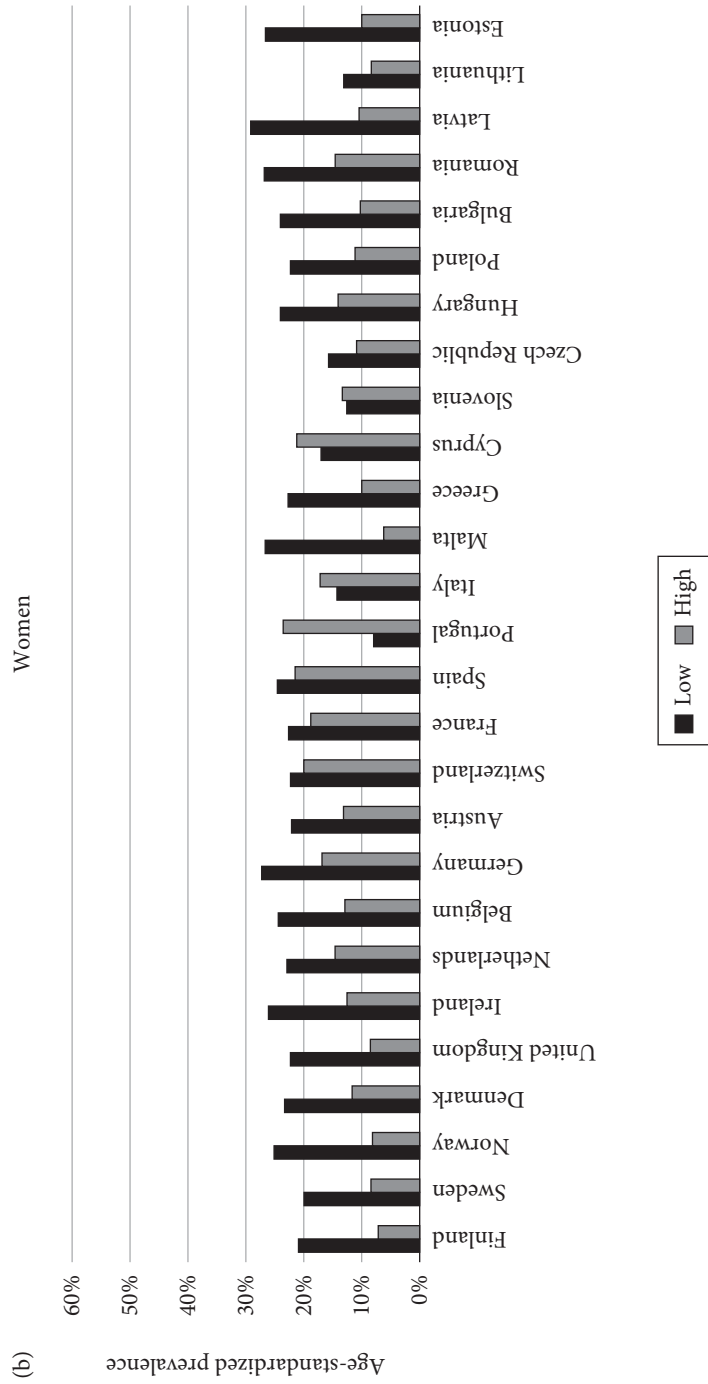


Figure 3.5 Educational inequalities in smoking

Notes: Smoking = current smoking. Data for ca. 2010. Average of 3 surveys, adjusted for average per survey.

Source: dataset constructed in DEMETRIQ/LIFEPAH projects, with harmonized data from European Health Interview Survey, European Social Survey, and National Health Interview Surveys.

Diet, physical activity, and obesity

Men and women in lower socioeconomic groups tend to eat less fresh vegetables and fruits, particularly in the North of Europe. Differences in fresh vegetable and fruit consumption are smaller in the South of Europe, probably because of the larger availability and affordability of vegetables and fruits in Mediterranean countries.⁴⁰⁶ Literature reviews have shown that it is likely that many other aspects of diet, such as consumption of meat, dairy products, and various fats and oils, also are socially patterned in many European countries, and that these social patterns differ between countries, reflecting long-standing differences in food culture.^{415,416}

Little or no leisure time physical activity tends to be more common in the lower socioeconomic groups. Survey data show that people with a higher education on average have a lot more physical exercise in their leisure time, for example in the form of sports activities. Although people with a lower education used to be more physically active during work, this is no longer the case today.^{417,418}

As a result of differences in food intake and physical exercise, overweight and obesity are usually more common in lower socioeconomic groups, but the magnitude (and sometimes even the direction) of these inequalities again differs strongly between countries.^{132,419} For example, among men absolute inequalities are small in Scotland and in Central-Eastern and Eastern Europe because of a very high prevalence among the higher educated, whereas among women inequalities are large in Southern Europe because of a very low prevalence among the higher educated (Figure 3.6).

Interestingly, this is one of the very few health aspects where socioeconomic inequalities are larger for women than for men. Women with a higher level of education are far less often overweight or obese than men with the same educational level, probably because they are more worried about their weight than boys and men from the same groups, and more actively try to manage their weight.¹³²

Mediation analyses show that inequalities in physical activity and obesity play a role in generating health inequalities,^{410,411,420,421} but the contribution of inequalities in obesity to inequalities in health differs strongly between countries.²⁶³

Excessive alcohol consumption

Excessive alcohol consumption is bad for health too, raising the risk of a wide range of health problems including various cancers, cardiovascular disease, liver cirrhosis, psychiatric disorders, and unintentional and intentional injuries.⁴²²

As in the case of smoking and obesity, the social patterning of harmful drinking differs between countries and between men and women. Whereas the association between socioeconomic position and *drinking status* is uniformly positive across all countries and genders (i.e. whether one drinks alcohol at all is more common in higher socioeconomic groups), the association between socioeconomic position and *harmful drinking* (either in the form of regular consumption of hazardous amounts exceeding recommended limits, or in the form of heavy episodic or 'binge' drinking) is very heterogeneous.^{423,424}

Among men drinking regular hazardous amounts is more frequent in lower socioeconomic groups in some countries (Switzerland, Spain, Hungary, ...), but more frequent in higher socioeconomic groups in other countries (Finland, England, Germany, ...).⁴²⁴ Similarly, although heavy episodic drinking is more frequent among men in lower socioeconomic groups in most countries, in some countries (England, France, Czech Republic) the reverse seems to be the case.^{423,424} Among women, drinking regular hazardous amounts is uniformly more frequent in higher socioeconomic groups, but heavy episodic drinking is more frequent in higher socioeconomic groups in some countries, and in lower socioeconomic groups in other countries.^{423,424}

At the same time, studies consistently show that alcohol-related problems, such as alcohol dependence, social and financial difficulties, and alcohol-related diseases and injuries, are more common in lower socioeconomic groups, even after taking alcohol consumption patterns into account. This may be due to the fact that the risks of harmful drinking can be buffered by a favourable social environment, for example, friends and relatives helping to prevent problems.^{257,425} Mediation analyses confirm that excessive alcohol consumption does contribute to the explanation of health inequalities in some countries, but less so in others.^{410,411,421}

Determinants of health-related behaviour

The systematic nature of these differences in health-related behaviour clearly demonstrates that these are not only a matter of free choice, but at least partly determined by constraints imposed by conditions beyond the control of the individual. **If smoking, eating fruits and vegetables, physical exercise, and excessive drinking were a matter of free choice, one would expect random patterns, not the systematic differences between socioeconomic groups that we have seen in the previous pages.**³⁷⁴

A lot of research, from various angles, has been done to try to understand why people in lower socioeconomic groups tend to have less healthy behaviour patterns. A recent review identified nine different causal mechanisms that have found empirical support.⁴²⁶

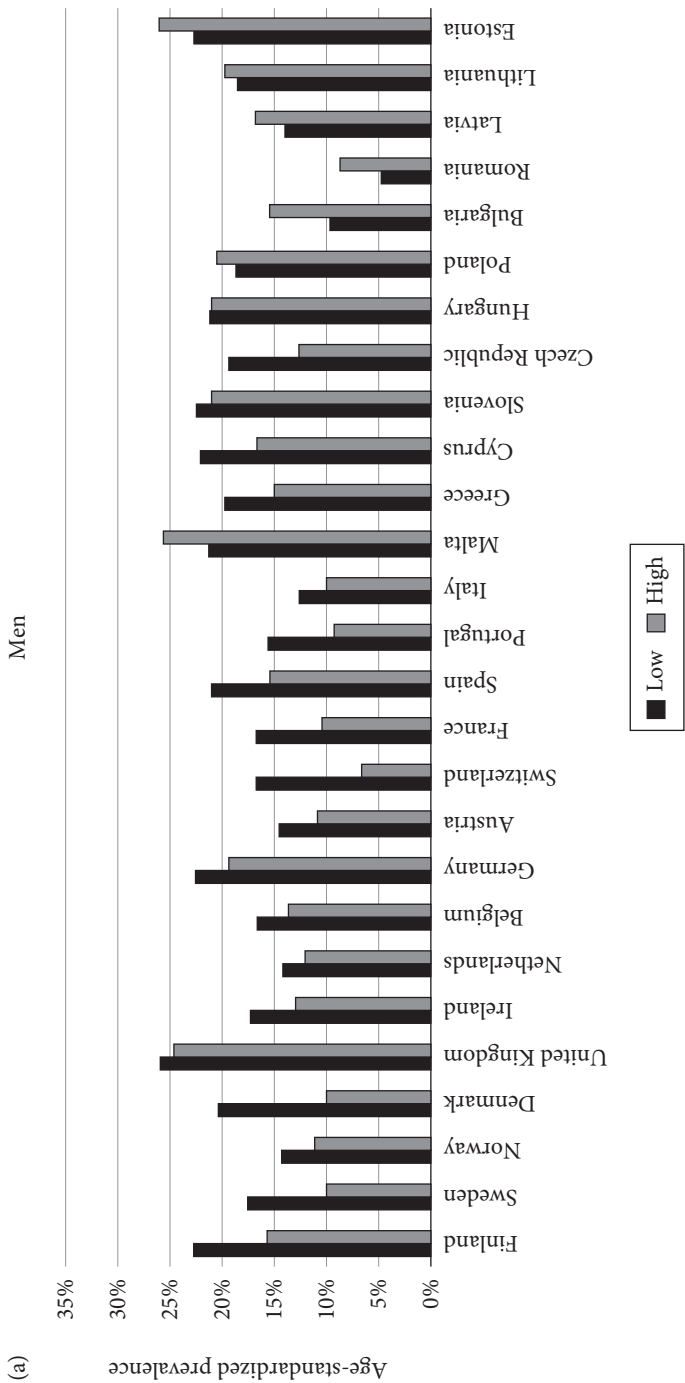
Four of these mechanisms are about incentives or motivations for healthy behaviour. The general idea is that people in lower socioeconomic groups have less reason to forego the short-term pleasures of unhealthy behaviour (smoking, sedentary living, over-eating, excessive drinking) in exchange for a long-term gain in longevity because: (i) **they face more psychosocial stress that encourages coping through engaging in unhealthy behaviour;** (ii) **they gain less longevity benefits from healthy behaviour, because they have lower life expectancy anyway;** (iii) **they have less need to gain prestige by setting themselves apart with healthy behaviour;** and (iv) **they have less 'health literacy', and thus knowledge of health risks.**⁴²⁶

Some other mechanisms are about the means to reach health goals. The general idea here is that, even if lower and higher socioeconomic groups would have a similar motivation for healthy behaviour, lower socioeconomic groups would still have more difficulty in realizing their goals, because: (v) they have less self-efficacy, problem-solving skills, ability to process information, and general sense of control needed to overcome obstacles for healthy behaviour; (vi) they have less financial resources for healthy behaviour, as in the case of costly healthy diets and sports; (vii) they have less opportunities in their neighbourhood for healthy behaviour, such as grocery stores with fresh fruits and vegetables and green spaces for leisure-time physical activity; and (viii) they receive less social support and less positive peer influence to promote healthy behaviours.⁴²⁶

A ninth mechanism affects both motivations and means: lower socioeconomic groups have (ix) latent traits, such as lower cognitive ability and certain personality characteristics, that are not conducive to healthy behaviour.⁴²⁶ To this we might add a tenth mechanism: (x) some of these inequalities in health behaviour may be due to reverse causation: as mentioned in section 3.1 'Methodological issues', obesity and excessive alcohol consumption may affect social mobility.

Health care

A final group of contributing factors is health care: if people with a lower socioeconomic position receive less, or lower quality, health care than people with a higher socioeconomic position, this exacerbates the inequalities in health generated by all the other contributing factors. Less use of effective preventive services by lower socioeconomic groups may contribute to a higher incidence or higher case fatality rate of diseases, and less use of effective treatment services may contribute



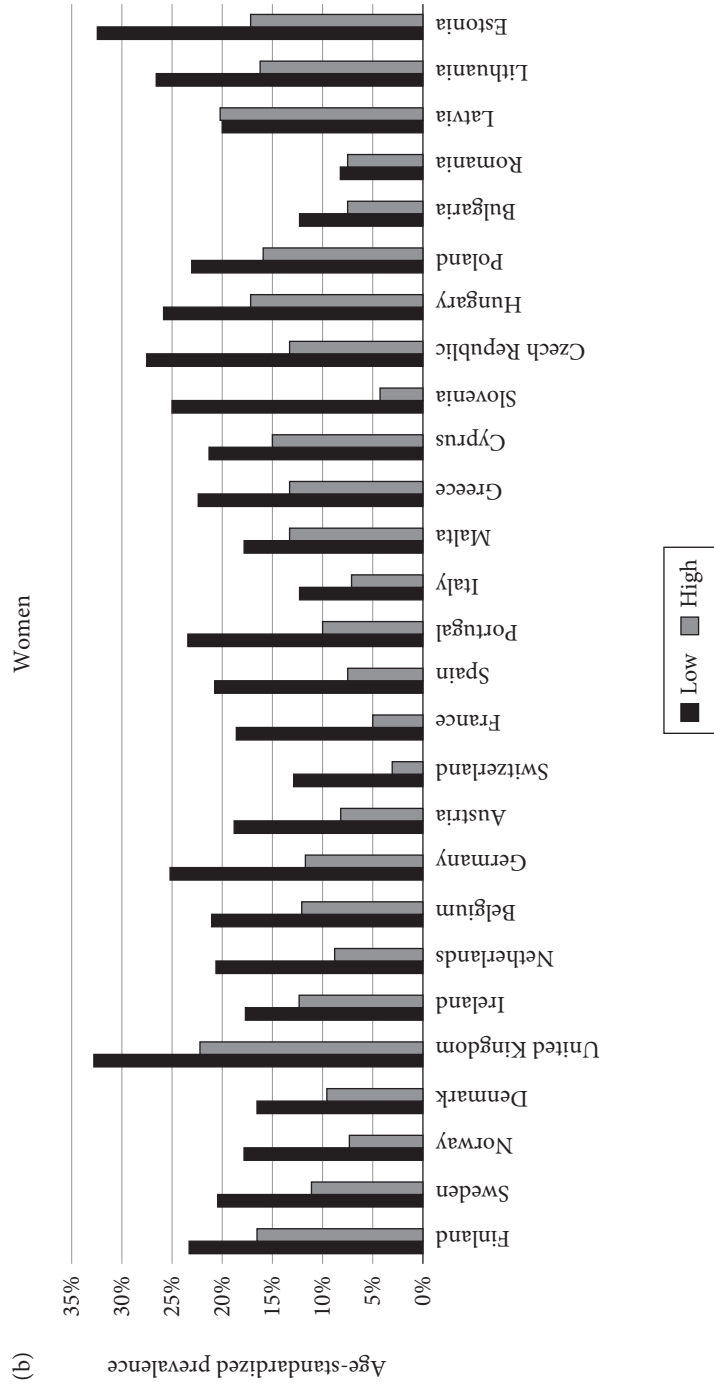


Figure 3.6 Educational inequalities in obesity

Notes: Obesity = Body mass index > 30 kg/m², calculated from self-reported weight and height. Data for ca. 2010. Average of 3 surveys, adjusted for average per survey.
Source: dataset constructed in DEMETRIQ/LIFEPAATH projects, with harmonized data European Health Interview Survey, European Social Survey, and National Health Interview Surveys.

to a lower recovery rate or higher case fatality rate of diseases. There are good reasons to suspect that this plays a role, depending on the degree to which countries have, or have not, eliminated inequalities in access and quality of health care.

Preventive health care

Prevention programmes, such as vaccinations, cardiovascular risk management, and cancer screening, can be very effective in improving population health. In order to avoid inequalities in health outcomes, it is important to implement them in such a way that lower socioeconomic groups participate at the same rate as higher socioeconomic groups, and receive the same quality of care. However, this is not always the case.

Many countries have cancer screening policies, particularly for cervical cancer, breast cancer and colorectal cancer. Of these breast cancer screening policies have been studied most extensively. More and more countries now have population-based breast cancer screening programmes, in which all members of the target group (e.g. women aged 50–75 years) are individually identified and invited to attend screening. However, opportunistic screening, in which invitations depend on the individual's decision to ask for screening or on encounters with health-care providers, has been more common in the past.⁴²⁷

Women from higher socioeconomic groups tend to have higher rates of participation in breast cancer screening, but these inequalities are smaller in countries with population-based screening than in countries with opportunistic screening or regional screening programmes.⁴²⁸ Within population-based screening programmes, variations in design also affect participation rates. Some designs enhance access to screening among lower socioeconomic groups, such as low cost (free tests and reducing geographical barriers), involvement of primary-care physicians, and individually tailored pro-active communication policies that address barriers to screening.⁴²⁹ It is therefore likely that some screening programmes, through inequalities in participation rates, contribute to a widening of inequalities in health outcomes.

Curative health care

Although most high-income countries have created financing systems that have substantially reduced financial barriers to health care use, these and other barriers have not been completely eliminated and still generate differences in health care use between socioeconomic groups, as shown by an important series of comparative studies.^{430–432} The most recent of these studies found 'horizontal inequities' in health care use (i.e. differences in health care use between income groups after adjustment for 'need', or health status) in all OECD countries. Whereas the poor are as likely to see a general practitioner as the rich, the rich are much more likely to see a specialist and a dentist, and to participate in cancer screening.⁴³³ Similar differences in doctor visits and use of a range of preventive services are found when education is used as an indicator of socioeconomic position.^{428,434,435}

Although these inequalities are found nearly everywhere, countries differ substantially in their magnitude. Some of these differences are related to health systems characteristics, particularly the extent of public health insurance coverage and the level of out-of-pocket payments for different services. Countries with less public, and more private insurance coverage have larger inequalities in doctor visits, and so do countries with higher out-of-pocket payments.⁴³³

That financial barriers may play an important role in generating inequalities in health care use, but to a differing extent in different countries, can be seen in Figure 3.7. This shows differences between low and high educated in 'unmet needs for medical care' for financial reasons. In many European countries, the proportion of people reporting such unmet needs is very low, even among people in the lowest income quintile. However, proportions exceeding 5% among those with a low

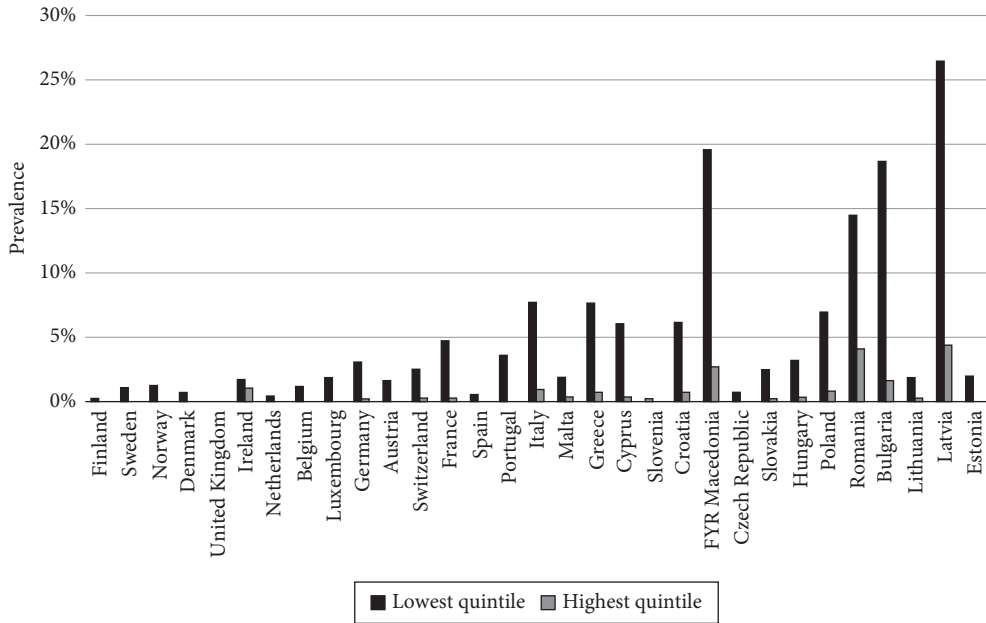


Figure 3.7 Inequalities by income in unmet need for medical care

Notes: Unmet need = self-reported assessment of whether respondent needed medical examination or treatment (dental care excluded), but did not have or seek it, because 'too expensive'. Income in quintiles of equivalized net household income. Data for 2010.

Source: data from EU Statistics on Income and Living Conditions via Eurostat (<http://ec.europa.eu/eurostat/data/database>; accessed 28 February 2018).

income and large inequalities between income groups are found in several Southern European countries (Italy, Greece, Cyprus) as well as in many Central-Eastern and Eastern European countries.⁴³⁶ Similar results are found for low versus high education.

Whether these inequalities in health care use generate inequalities in health outcomes depends on the effectiveness of the specific interventions that the services deliver, and are forgone by those who do not use the service. Two separate pieces of evidence suggest that this is indeed the case. First, for many specific interventions inequalities in use favouring people with a higher socioeconomic position have been found in many European countries. These include interventions of proven effectiveness: cardiac revascularization (by coronary bypass grafting or percutaneous coronary intervention),⁴³⁷ common surgical procedures,⁴³⁸ statins,⁴³⁹ various treatments for lung and colorectal cancer,^{139,440} diagnosis and control of diabetes mellitus,⁴⁴¹ and many others.

Despite low levels and small inequalities in unmet need (Figure 3.7), these inequalities have also been found in Finland, where they can be studied in detail thanks to the availability of linked administrative databases. The Finnish experience suggests that it is reasonable to expect other Western European countries to have similar inequalities in use of specific therapeutic interventions, and to expect many Southern, Central-Eastern and Eastern European countries to have even larger inequalities in use of specific therapeutic interventions.

Inequalities in health care outcomes

A second type of evidence suggesting that inequalities in health care use generate inequalities in health outcomes is that inequalities in case fatality rates have been found for many diseases,

and that these are likely due, at least partly, to inequalities in treatment. European studies of socioeconomic inequalities in survival after myocardial infarction have often found short- and long-term case fatality after the event to be higher among patients with lower socioeconomic position.^{135–137} Also, studies of inequalities in cancer survival have almost universally shown case fatality to be higher among patients with lower socioeconomic status^{138–140}—a finding that can only partly be explained by inequalities in diagnosis stage or other patient or tumour characteristics, suggesting that inequalities in treatment must also play a role.⁴⁴²

Whereas these two pieces of evidence clearly indicate that inequalities in health care play a role in exacerbating inequalities in health, particularly in countries where inequalities in health care access and utilization are large, it is unknown what their quantitative contribution is. Formal mediation analyses are rare and fraught with methodological problems (such as incomplete measurement of non-health care determinants of survival and confounding by indication⁴⁴³). The most suggestive (but still indirect) evidence comes from studies of inequalities in mortality from conditions amenable to medical intervention, such as cerebrovascular disease, tuberculosis, appendicitis and perinatal mortality (see Chapter 4).

Biological mechanisms

Health inequalities can thus be thought of as resulting from differences between socioeconomic groups in a wide range of contributing factors, playing out over the life-course: genetics, childhood environment, material living conditions, social and psychological factors, health-related behaviours, and health care. But before these factors can cause disease, they have to ‘get under the skin,’ to use a popular expression among social epidemiologists.

Embodiment

American social epidemiologist Nancy Krieger has introduced the concept of ‘embodiment’—the notion that ‘we [human beings] literally embody, biologically, our lived experience’, that ‘our living bodies tell stories about our lives, whether or not these are ever consciously expressed’, and that differences in health status between population groups are the biological expression of social relations.⁴⁴⁴ (p. 350) This is a powerful idea that can potentially capture a wide variety of pathways linking the social to the biological, including the effects of exposure to toxic substances in the work environment, tobacco smoking, major life events, and many other ‘social-biological transitions’.⁴⁴⁵

Attention to these biological pathways has increased importantly with the rise of interest in biomarkers, that is, indicators of some biological state or condition that measure exposure to determinants of disease or early harmful effects of such exposures on the body. The rise of interest in biomarkers has been propelled by technological developments in areas such as molecular biology and imaging, which have greatly expanded the scope for measuring the physiological state of the human body.

There must be many different biological pathways linking lower socioeconomic position to ill-health and premature death, but it may well be that some of these pathways have a more generic effect, in the sense that they link several different health outcomes to lower socioeconomic position. The ‘embodiment’ of psychosocial stress is an important candidate for such a more generic mechanism.

The biological response to psychosocial stress

As mentioned above, the experience of stress affects the neural, endocrine, and immune systems of the body, and chronic stress may lead to maladaptive responses in the form of, for example,

high blood pressure, a prolonged high level of cortisol, higher blood viscosity, or a suppression of the immune response, which may in their turn increase the susceptibility to a range of diseases.^{398–400}

The **neuroendocrine response to psychosocial stressors** comes in the form of a fast response mediated by the **sympathoadrenal pathway**, and a slow response mediated by the hypothalamic-pituitary-adrenal axis. As a result of the first, **epinephrine (adrenalin) is released into the bloodstream; as a result of the second, cortisol is released into the bloodstream.** This biological stress response is adaptive, but if it is activated for too long or too often it may have deleterious health effects. Because people from lower socioeconomic strata are more exposed to psychosocial stressors, they may suffer more from these deleterious health effects.^{399,400}

Both epinephrine and cortisol have a wide range of effects on metabolism, the haemostatic and cardiovascular systems, immunity, and cognitive functioning. These effects have been documented in animal studies, and to some extent in human studies, but it is not yet clear what the quantitative contribution of such mechanisms is to the explanation of inequalities in health. Evidence encompassing complete pathways from socioeconomic status through neuroendocrine responses to ill-health is still rare.⁴⁴⁶

Some of it comes from the British Whitehall study, which found strong socioeconomic gradients in prevalence of the 'metabolic syndrome', that is, a combination of impaired glucose tolerance, lipid abnormalities, and central obesity. These gradients could only partly be explained by conventional risk factors such as smoking, obesity, and other behavioural factors, and were partly explained by socioeconomic differences in exposure to job stress. **This suggests that there is a separate biological pathway running from low socioeconomic position through psychosocial stress and metabolic factors to heart disease, diabetes, and other chronic conditions.**⁴⁴⁶

Allostatic load

The idea that chronic life stress, in various forms, leads to physiological dysregulation and thereby to poor mental and physical health has been more fully developed in the theory of 'allostasis' and 'allostatic load'. **'Allostasis' is the adaptive process that maintains stability or 'homeostasis' of the body during exposure to physical and psychological stressors, and while this is critical to survival it also carries a cost, that is, 'allostatic load' which is defined as the 'cumulative physiological burden exacted on the body through attempts to adapt to life's demands.'**⁴⁴⁷

This is commonly operationalized with a battery of biological parameters, not only including indicators of neuroendocrine activity (e.g. epinephrine, cortisol) but also metabolic factors (e.g. cholesterol, glucose, obesity), cardiovascular factors (e.g. blood pressure, heart rate), and biomarkers relating to the immune system and inflammatory processes (e.g. interleukins, C-reactive protein).^{448,449} Cumulative scores of these biomarkers appear to predict mortality, as well as declines in cognitive and physical functioning, independent of other factors.⁴⁴⁹

Socioeconomic gradients in allostatic load have been found in many countries, **with lower socioeconomic groups having higher cumulative scores.**⁴⁴⁸ These gradients, together with the general rise of allostatic load with age, have been interpreted as indicating differences between socioeconomic groups in 'wear and tear' of their bodies, **with lower socioeconomic groups accumulating allostatic load over their life course at a higher rate than lower socioeconomic groups.**⁴⁵⁰ In one study from the United States, **allostatic load explained about a third of educational inequalities in mortality.**⁴⁵¹ Of course, socioeconomic inequalities in allostatic load are themselves a result of differences between socioeconomic groups in material, psychosocial and behavioural factors.⁴⁵²

3.4 Theories about the explanation of health inequalities

Scientific explanation

The available evidence thus suggests that health inequalities can partly be explained by differential exposure over the life-course to a range of well-known health determinants, and that these factors are ultimately ‘embodied’ by various biological pathways. It would be useful if these insights could be integrated into a general scientific theory of how health inequalities arise, for example along the lines of the famous ‘covering law model’ of German philosopher Carl Hempel (1905–97). He proposed that scientific explanation should consist of creating a logical argument leading from a set of ‘general laws’ and ‘initial conditions’ to the ‘phenomenon to be explained’. For example: Newton’s general law of gravitation and the fact that the apple is cut loose from the tree, together explain why the apple falls towards the ground.⁴⁵³

This model suggests that we should try to find one or more ‘general laws’ and ‘initial conditions’ that help us to understand why people with a lower socioeconomic position tend to have more health problems and to die at an earlier age than people with a higher socioeconomic position. To give an example that we will encounter in a more elaborate form later on: health inequalities (the ‘phenomenon to be explained’) arise because all human beings are evolutionarily programmed to avoid illness and death (the ‘general law’) and because people with a lower socioeconomic position have less means of avoiding illness and death (the ‘initial conditions’).

In this section we will review a number of attempts at such a more general understanding of how health inequalities arise, including why they are so persistent. Although a single, comprehensive, and commonly accepted scientific theory of health inequalities does not exist, a number of theories have been proposed that try to explain health inequalities at a more general level of understanding than in terms of the distribution of a shorter or longer list of contributing factors (Table 3.1). None of these theories is mutually exclusive, and it may well be that several, or even all of them, capture a relevant part of reality.

The theories have been classified in four groups, depending on their main focus: selection into socioeconomic groups, differential health progress, social disadvantage, and social production of disease. The order in which the theories are presented is not entirely arbitrary: in this order they go from a more ‘benign’ attribution of health inequalities to social selection (e.g. ‘health inequalities are mainly the result of differences in composition of socioeconomic groups, and have little or nothing to do with the extent of social inequality in a society’) to a more ‘conflictual’ attribution to the social production of disease (e.g. ‘health inequalities ultimately result from abuses of power and privilege by the higher socioeconomic groups’).

In the column labelled ‘proposed explanation for health inequalities’ we have tried to summarize each theory in three statements: something resembling a general law (e.g. ‘medical care helps to lower morbidity and mortality’), one or more initial conditions (e.g. ‘due to market forces the supply of medical care is inversely proportional to need’) and the phenomenon to be explained (, [as a result,] higher socioeconomic groups have lower morbidity and mortality’).

Theory focusing on selection into socioeconomic groups

The main thrust of this theory is that health inequalities are at least partly due to selection factors, either in the form of health-related selection or reverse causation (i.e. health determining entry into and/or exit from socioeconomic groups) or ‘indirect selection’ or confounding (i.e. personal characteristics such as cognitive ability and personality traits determining entry into and/or exit from socioeconomic groups).

As we saw in section 3.2 ‘Education, occupation, income, and health’, these mechanisms do plausibly play a role in generating health inequalities by education, occupational class, and income,

Table 3.1 General theories explaining health inequalities

Group	Name of theory	Scientific origins, proponent(s), selected references	Proposed explanation for health inequalities
I. Selection into socioeconomic groups	Social selection	Michael Young ⁵⁷ Patrick West ²³⁰ Johan Mackenbach ²³⁶	Personal characteristics like cognitive ability and personality traits determine whether people can effectively use opportunities for health improvement. In modern societies, people are socially mobile, and 'sorted' into socioeconomic groups based on their personal characteristics. As a result, higher socioeconomic groups have better health.
II. Differential health progress	Diffusion of innovations	Eugene Rogers ⁴⁵⁴ Alan Lopez ⁴⁵⁵	Health improvements often result from the adoption of new behaviours, and due to their tendency for earlier uptake of new ideas higher socioeconomic groups are usually the first to adopt new behaviours. As a result, the higher socioeconomic groups are also the first to benefit from most health improvements.
	Cultural capital	Pierre Bourdieu ⁴⁵⁶ Thomas Abel ⁴⁵⁷	Cultural resources, such as beliefs, preferences and knowledge, affect health because health improvement is driven by lifestyle changes, and inequalities in such 'cultural capital' are still large in modern societies' socioeconomic groups. As a result, higher socioeconomic groups have better health.
	Inverse equity	Cesar Victora ⁴⁵⁸	Health improvements often result from the introduction of new interventions, and due to a relative absence of cultural, geographical and financial barriers to the uptake of new interventions higher socioeconomic groups are usually the first to use them. As a result, the higher socioeconomic groups are also the first to benefit from most health improvements.
III. Social disadvantage	Inverse care law	Julian Tudor Hart ⁴⁵⁹	Medical care helps to lower morbidity and mortality, but due to market forces the supply of medical care is still inversely proportional to need, i.e. lower in more disadvantaged areas. As a result, higher socioeconomic groups have lower morbidity and mortality.

(continued)

↑ CLASSES = ACESSO CULTURA, INOVAÇÃO E NOVOS TRATAMENTOS

Table3.1 Continued

Group	Name of theory	Scientific origins, proponent(s), selected references	Proposed explanation for health inequalities
	Neo-materialism	Black Report ²⁰ John Lynch ⁴⁶⁰ George Davey Smith ⁴⁶¹	Material resources, both at the individual and community level, are important for health, and due to the way material resources are distributed in society higher socioeconomic groups still have more of them. As a result, higher socioeconomic groups also accumulate less health-damaging exposures and experience over their life-course and have better health.
	Psychosocial environment	John Cassel ⁴⁶² Michael Marmot ⁴⁶³ Richard Wilkinson ⁴⁶⁴	A lower relative position in society, and the perception of the associated inequalities in prestige and access to resources, affects health by giving rise to psychosocial stress, and differences in relative social position are still large in modern societies. As a result, higher socioeconomic groups have better health.
IV. Social production of disease	Fundamental causes	Stanley Lieberman ⁴⁶⁵ Bruce Link ⁴⁶⁶ Jo Phelan ⁴⁶⁷	A higher socioeconomic position provides the individual with flexible resources, such as knowledge, money, power, prestige, and beneficial social connections, which can be used to avoid disease risks or to minimize the consequences of disease once it occurs, regardless of the prevailing disease and risk factor profile. As a result, higher socioeconomic groups have better health.
	Political economy of health	Lesley Doyal ⁴⁶⁸ Vicente Navarro ⁴⁶⁹ Nancy Krieger ⁴⁷⁰	Inequalities in political and economic power in society produce inequalities in the material and social conditions under which people live and work, and these conditions structure their exposure to health risks and opportunities for health improvement. As a result, higher socioeconomic groups have better health.

and what this theory claims is that trends and differences in the magnitude of health inequalities can also be understood in part from trends and differences in the rates of social mobility, or in the degree of selectivity on health or personal characteristics of social mobility, or both. To be more precise: that more social mobility, and more selectivity during social mobility, leads to larger health inequalities, all other things being equal.

The idea originates with British medical sociologist Patrick West, who ‘re-thinking the health selection explanation for health inequalities’ emphasized the important role of education in determining intergenerational social mobility, and proposed a central role in the generation of health inequalities for the ‘attitudes, experiences or behaviours [...] associated with different

[educational] aspirations or orientations which [...] have consequences for future mobility trajectories [and] are correlated with health.²³⁰ (p. 380)

Social mobility and health inequalities

Building on these ideas, it has been proposed that selection during intergenerational social mobility might explain the persistence of health inequalities over time despite the inequalities-dampening effects of the welfare state, because intergenerational social mobility has increased over time, and has probably become more selective with regard to health-relevant personal characteristics, particularly in countries with generous welfare arrangements.¹³

Trends in social mobility have been studied extensively, and there is no doubt that intergenerational social mobility has increased in all European countries, in the sense that birth cohorts born around the middle of the twentieth century have been more socially mobile, particularly upwardly mobile, than earlier born birth cohorts. This applies to both education and occupational class, and is the result, quite simply, of changes in the overall distribution of education and occupational class in the population.⁴⁷¹

Due to educational expansion and economic modernization, the proportion of people with a higher level of education and a non-manual occupational class has increased in all European populations, and this upward shift in educational and occupational levels has implied, and even required, substantial upward mobility of children as compared to their parents. Stratification sociologists call this 'absolute mobility', and it has increased between birth cohorts born in the beginning, and those born around the middle of the twentieth century in all European countries.⁴⁷¹

It is also likely that intergenerational social mobility has become more selective with regard to individual characteristics that are relevant for health, but here the evidence is only indirect, and based on the weakening association between the socioeconomic position of parents and that of their children. It used to be the case that parents' socioeconomic position (in terms of education, occupational class, income) was a strong predictor of their children's socioeconomic position. This leaves little scope for selection on children's personal characteristics that may predispose them for illness later in their lives, or will keep them healthy.

However, this association has weakened over time, as is clear from studies which have shown that 'relative mobility', or 'social fluidity' (i.e. upward or downward mobility net of changes in the overall distribution of education and occupational class in the population) has increased over time. Although a famous study by Swedish sociologist Robert Erikson and John Goldthorpe found relative mobility rates to be more or less constant over time,⁵⁹ a more recent and more detailed study by British sociologist Richard Breen found an increase in social fluidity, in terms of both educational and occupational class mobility, for birth cohorts born between the 1920s and 1960s in many European countries (with stagnation thereafter).⁴⁷² This was probably facilitated by structural changes in the economy and educational expansion, but also helped by educational reforms aiming at an equalization of opportunities for higher education.⁴⁷³

Whereas trends in intergenerational social mobility may partly explain the persistence of health inequalities over time, differences between countries in social mobility may partly explain why some countries have larger health inequalities than others. Relative intergenerational mobility—whether classified by education, occupational class or income—has been lower-than-average in Southern Europe, France, and Germany, and higher-than-average in the Nordic countries, the Netherlands, and (post-)communist countries in Central-Eastern and Eastern Europe, with the United Kingdom in-between.^{474,475} This may help to understand why the Nordic countries have larger health inequalities than expected on the basis of the magnitude of their inequalities in poverty and other material living conditions.⁴⁷⁶ We will further investigate this mechanism in Chapter 4.

Meritocracy

To some extent, modern societies have become ‘meritocracies’, in which a person’s educational and occupational achievement is no longer dependent on family background but on his or her own talents and efforts, and thus on personal characteristics like cognitive ability and personality.⁴⁷⁷ Because these personal characteristics are also important for health, for example, because they influence health-related behaviour, inequalities in health probably have increased as a consequence. The tragedy thus is that in modern societies social inequality has found new ways to sustain and reproduce itself, and that old forms of disadvantage have partly been replaced by new forms of disadvantage, in the disguise of personal characteristics like cognitive ability and personality.

It is important to remind ourselves that the term ‘meritocracy’ was coined by British sociologist Michael Young (1915–2002), not to *celebrate* but to *criticize* the increasing dependence of educational achievement on intelligence, and the emergence of a new elite based on very narrow selection criteria.⁵⁷ However, although European societies have indeed become more meritocratic, the association between parental socioeconomic position and children’s socioeconomic positions has not disappeared, and **there is still intergenerational transmission of earnings and wealth, due to cultural transmission of savings behaviour, genetic inheritance of cognitive skills and other determinants of economic success, and direct bequests of money.**⁴⁷⁸

Also, and paradoxically, educational achievement now explains more of the association between social origin and occupational class than in the past, due to the fact that parents in higher occupational classes are better able to let their children succeed in the educational system. In this way, meritocratization is actually amplifying inequality in occupational position by social origin.⁴⁷²

Theories focusing on differential health progress

The main thrust of these theories is that the **persistence of health inequalities is due to the fact that health is continually improving, but at different speeds** in lower and higher socioeconomic groups. According to American economist and Nobel laureate Angus Deaton ‘inequality is the handmaiden of progress’, and both economic and health progress are often accompanied by a widening of inequalities, both between and within countries.⁴⁷⁹ The ‘diffusion of innovations’, ‘cultural capital’, and ‘inverse equity’ theories offer different but overlapping explanations for this phenomenon.

Diffusion of innovations

Studies of the **‘diffusion of innovations’ have found that people with a higher socioeconomic position often are early adopters of new behaviours in many fields, only later to be followed by people with a lower social position.**⁴⁵⁴ This also applies to many health-related behaviours, such as cigarette smoking. The higher socioeconomic groups were the first to take up the new habit, but also the first to stop, and as a result the declining phase of the smoking epidemic is often accompanied by widening inequalities in smoking. A ‘model of the cigarette epidemic’ has been proposed which describes the diffusion of smoking behaviour through populations in four stages.⁴⁵⁵

In stage 1, smoking is an exceptional behaviour and mainly a habit of men, particularly from higher socioeconomic groups. In stage 2, smoking becomes ever more common, and is now also picked up by women in higher socioeconomic groups and by men in lower socioeconomic groups. In stage 3, prevalence rates among men start to decrease since many men stop smoking, especially those who are better off, but women reach their peak rate only later during this stage.⁴⁵⁵ In stage 4, smoking rates keep declining for both men and women, and smoking becomes concentrated in lower socioeconomic groups.^x

This descriptive model more or less fits the European experience, including the reversal from a positive to a negative association between socioeconomic status and smoking found in many

European countries.^{413,414,480} The underlying mechanism may indeed be a delay in the diffusion of new behaviour, but other factors, such as lack of effectiveness of health education efforts among lower socioeconomic groups, may also have played a role.

In the case of obesity the higher socioeconomic groups also historically were the first to become overweight, but as a clear decline has not yet set in we do not know whether a similar widening will occur.¹³² In some developing countries, obesity is still more common among people of higher socioeconomic position, and with economic development the burden of overweight gradually shifts towards lower socioeconomic groups.⁴⁸¹ This illustrates that processes of diffusion of behaviour, such as dietary habits and physical exercise, may play a role here too, but that these are conditioned by the availability of food and the necessity of physical exercise.

Cultural capital

Another approach to the understanding of behaviour, and behaviour change, is to acknowledge the fact that behaviour, whether healthy or unhealthy, can act as a status symbol: the choice of a particular 'lifestyle' signals to others that one belongs to a particular social class. This may apply to eating habits, language use, leisure activities, musical preferences, and many other aspects of everyday culture.

French sociologist Pierre Bourdieu (1930–2002) has given a classic description of these differences in his book 'Distinction; a social critique of the judgement of taste', and was also the first to apply this idea to preferences with regard to body shape and size. **He found that people with a higher socioeconomic position were more likely to consume lean and light food products, and viewed thinness as an image of success and willpower, whereas lower socioeconomic groups were more concerned with the strength of the body.** He even saw the human body as 'the most indisputable materialization of class taste'.⁴⁵⁶ (p. 190)

Many differences in behaviour, and underlying differences in the social prestige of behaviour, show a distinct dynamic. **Aspects of 'lifestyle' in modern societies are often first adopted by the higher social classes, and only later become more widely distributed.** At that moment, however, they lose their distinctive character, and it becomes attractive for the higher social classes to abolish the behaviour and/or to search for new sources of distinction. This phenomenon is also known under various other names, such as 'sinking cultural goods' (a term popular in continental Europe that was introduced by German folklorist Hans Naumann (1886–1951)) and 'trickle-down effect' (a concept more popular in the English-speaking world).⁴⁸²

For example, until the end of the nineteenth century, eating too much and exercising too little were reserved for the wealthy, but they were taken over by the less privileged when food became available in abundance and the necessity for physical labour disappeared. The well-to-do then found a new source of distinction in a new cultural norm of paying attention to their weight and trying to remain slim.¹³²

People from different socio-economic groups also differ in what they find 'tasty': the taste preferences of people from lower socio-economic groups have been characterized as a 'taste of necessity', instead of a 'taste of luxury'.⁴⁸³ The higher socioeconomic groups try to distinguish themselves by emphasizing the health aspect in their diet, while the lower socioeconomic groups tend to attach more value to recently acquired luxury products such as white bread, meat, sweets, and snacks. This is not only a confirmation of Bourdieu's theories, but also a reminder of times that food abundance was not yet normal.^{xi}

Such cultural differences also apply to preferences in body size, particularly among women who are more subject to these aesthetic standards than men.¹³² The interaction between cultural differences and sex-specific aesthetic norms may explain why being overweight among women

shows a stronger correlation with socio-economic status than among men (section 3.3 'Six groups of contributing factors').

Differences between socioeconomic groups in preferences do not only depend on their economic resources, but may also depend on their cultural resources or 'cultural capital'. The concept of **cultural capital was introduced by Bourdieu to denote 'widely shared, high status cultural signals (attitudes, preferences, formal knowledge, behaviours, goods, and credentials) used for social and cultural exclusion'**.⁴⁸⁴

In the context of health inequalities, this theory explains inequalities in health-related behaviour from differences between socioeconomic groups in attitude, knowledge, and competency which are transmitted from one generation to the next.⁴⁵⁷ Empirical studies suggest that differences in cultural capital indeed partly explain socioeconomic inequalities in dietary patterns,⁴⁸⁵ physical activity,⁴⁸⁶ and body weight.⁴⁸⁷

Applying this perspective to the paradox of persisting health inequalities, two questions arise. Did the welfare state reduce inequalities in cultural capital? This seems unlikely,⁴⁸⁸ and persistence of inequalities in cultural capital may then partly explain persistence of health inequalities. Did the welfare state have an effect on the need for 'social distinction' on the basis of health-related behaviour? Well, it may have had. Due to the decline in opportunities to distinguish oneself by outward signs of material prosperity, the need for distinction on the basis of non-material signals like leisure-time physical exercise, a slim body or eating quinoa may well have increased.

Inverse equity

The 'inverse equity' hypothesis postulates that **'new interventions will initially reach those of higher socioeconomic status and only later affect the poor' [...]** which results in an early increase in inequity ratios.⁴⁵⁸ It predicts that after introduction of a new intervention, if there are cultural, geographical, or financial barriers to access, health inequalities will first increase and then decline when the intervention gradually reaches the most deprived groups, after the coverage among the most privileged groups has already reached 100%.⁴⁵⁸

It is based on a line of reasoning similar to that of the diffusion of innovations theory described above, **but focuses on the emergence of inequalities in the use of preventive or curative health care during periods of introduction of new interventions or services, which may be due to differences in both access and uptake.** Since its publication in 2000 this hypothesis has frequently been tested, mostly but not exclusively in low- and middle-income settings, and mostly with positive results.⁴⁸⁹

For example, in a worldwide analysis of the increase in hospital (instead of home) births, the magnitude of inequalities in hospital deliveries followed a striking curvilinear pattern: up to a national coverage rate of 50% socioeconomic inequalities increased, because of early adoption by the wealthy, and after this point had been passed inequalities declined again, because the poor also began to give birth in hospitals.⁴⁸⁹

The inverse equity hypothesis has less often been tested in a high-income setting, but it is easy to imagine that a combination of delayed access to new preventive and curative health care interventions, and delayed adoption of new health-promoting behaviours, partly explains inequalities in health improvements seen in so many European countries over the past decades.

Theories focusing on social disadvantage

The main thrust of theories focusing on social disadvantage is that health inequalities reflect, in a straightforward manner, inequalities in general living conditions which determine exposure to a wide range of disease risks, and that the persistence of health inequalities is due to the persistence

of inequalities in living conditions. The theories differ, however, in the sort of living conditions which they claim to be the most important.

The 'inverse care law'

When in the United Kingdom it became obvious that the National Health Service had not eliminated inequalities in mortality by occupational class, and people started to search for possible explanations, it soon became clear that inequalities in health care utilization had not disappeared at all. On the contrary, 'higher income groups know how to make better use of the service; they tend to receive more specialist attention; occupy more of the beds in better equipped and staffed hospitals; receive more elective surgery' etc.⁴⁹⁰ (cited in 459, p. 405)

Although financial barriers to access had largely been removed, the availability of good medical care differed importantly between areas, with general practitioners in more disadvantaged areas where most sickness and death occurs 'having more work, larger lists, less hospital support, [...] and hospital doctors shoulder[ing] heavier case-loads with fewer staff and equipment, more obsolete buildings, and suffer[ing] recurrent crises in the availability of beds and replacement staff'. This has since become known as the 'inverse care law': due to the fact that market forces still operate even in a National Health Service, 'the availability of good medical care tends to vary inversely with the need of the population served.'⁴⁵⁹ (p. 412)

This idea was a source of inspiration for the 'inverse equity' hypothesis described above, and could be generalized to include all services and benefits provided by the welfare state. Even when services and benefits are targeted to lower socioeconomic groups, those who are relatively better off, and thus are in less need, usually find it more easy to access the services and receive the benefits.

Neo-materialism

On various occasions we have already mentioned the Black report.²⁰ The writers of this report not only discussed the relative merits of 'causation' and 'selection' explanations, but also discussed the relative importance of various specific explanations, particularly behavioural and 'materialist' factors, concluding that the latter were the most important.²⁰

The idea that material living conditions are still important determinants of health, and important contributors to health inequalities, has sometimes been called a 'neo-material' or 'neo-materialist' interpretation.⁴⁶⁰ In contrast to explanations of health inequalities in behavioural or psychosocial terms, neo-materialist explanations emphasize the role of negative exposures (such as occupational risks and unfavourable housing conditions) and lack of resources (such as money and power) held by individuals, as well as the macro-level determinants of material living conditions (such as underinvestment in public infrastructure and services) in generating health inequalities.⁴⁶⁰

Many of the factors contributing to health inequalities that we reviewed in the previous paragraph can be seen as at least partly determined by the material living conditions of people with lower and higher socioeconomic positions.

Psychosocial environment

Neo-materialist explanations were partly emphasized as a critique on psychosocial explanations, which during the 1990s and early 2000s had a lot of traction in the scientific literature. One strand of research, described in section 3.3 'Six groups of contributing factors', focused on the role of psychosocial stress and the behavioural and biological mechanisms converting stress into ill-health and premature death.

Another strand of research, already briefly mentioned in section 1.3 'The need for a broader picture', focused on the association between income inequality and average population health

which was brought to the attention of a wide readership, both professional and lay, by Richard Wilkinson.^{40,41,464} In the course of his research and writing Wilkinson has offered several possible explanations for this relationship, but in his widely sold book ‘The Spirit Level; why greater equality makes societies stronger’ as well as in his most recent book ‘The inner level’ (both written with Kate Pickett) his main explanations are psychosocial.^{41,491}

In his view, **wide income inequalities, and wide social inequalities more generally, lead to loss of self-esteem and social insecurity among the worse off, as well as to feelings of inferiority and shame, to status anxiety, and loss of social trust.** A person’s relative position in the hierarchy of society has a crucial effect on his or her psychosocial well-being, and these psychosocial effects may then generate a wide variety of health effects, from illicit drug use to obesity, and from mental ill-health to premature death.^{41,491}

There is indeed some evidence that ‘status anxiety’, i.e. the degree to which people are concerned with their social status and with status competition, is higher in countries with more income inequality, not only among people with lower incomes but also among people with higher incomes,⁴⁹² which—as Wilkinson and Pickett speculate—could lead to higher levels of stress and a range of unfavourable health outcomes throughout society.⁴⁹¹

In a similar vein of reasoning, but with a stronger focus on health inequalities, Michael Marmot in his book ‘Status Syndrome; how social standing affects our health and longevity’, has hypothesized that the main mechanism linking lower socioeconomic position to ill-health and premature death is lack of control. In his view, how much control people have over their lives is crucial for their well-being and longevity, and it is the inequality in degrees of autonomy, power, and control that ultimately generates health inequalities.⁴⁶³

Empirical studies have found some support for these psychosocial theories. For example, a lower perceived social rank is associated with higher allostatic load,⁴⁹³ and inequalities in (perceived) control explain part of inequalities in mortality.⁴⁹⁴ **These psychosocial theories may also help to explain the persistence of health inequalities in modern welfare states: it is easy to imagine that the welfare state has been more effective in reducing inequalities in material living conditions, than in reducing inequalities in status and power within societies.**

Theories focusing on the social production of disease

Theories of ‘social production of disease’ come in different forms, but in common they emphasize the social origin of health inequalities. Instead of focusing on the differences in exposure, or behavior, or care access between people with a lower and higher socioeconomic position, they focus on how these differences come about. These inequalities in exposure, or behavior, or care access are not regarded as natural phenomena, but are seen as ‘produced’ by human action.⁴⁷⁰

Fundamental causes

A popular, but very general, framework for integrating much of the evidence on the explanation of health inequalities is the ‘fundamental causes’ theory, proposed by American social epidemiologists Bruce Link and Jo Phelan. This theory is based on the idea that **the social forces underlying social stratification are the ultimate causes of health inequalities, and that these provide a better explanation for health inequalities and their persistence, than differences in proximal risk factors such as smoking, working conditions, stressful life events, etc.** The general idea is that as long as social inequality persists, health inequality will also persist.

According to the fundamental causes theory the persistence of health inequalities over time and across different national contexts is due to the fact that a higher socioeconomic status provides people with ‘flexible resources’. These include ‘knowledge, money, power, prestige, and beneficial

social connections' which can be used 'to avoid disease risks or to minimize the consequences of disease once it occurs' regardless of the prevailing circumstances.⁴⁶⁶ (p. 487) The association between socioeconomic status and health then 'is reproduced over time via the replacement of intervening mechanisms', and when opportunities for avoiding or treating disease continue to expand, as they have consistently done over the past decades, health inequalities will continue to exist.⁴⁶⁷ (p. 629)

The fundamental causes theory thus gives a central role to the 'actionability' of health problems, and to purposive action or 'health-directed human agency'. Socioeconomic differences in the availability of the means to take advantage of this 'actionability' (by behaviour change, accessing health care, getting protection from occupational risks, applying road safety measures, ...) are the crucial factor on which the fundamental relationship between socioeconomic status and health rests.⁴⁹⁵

The mechanism posited in this theory thus emphasizes the role of the individual who disposes (or not, in the lower socioeconomic groups) over material and non-material resources which help him or her to protect and promote health. The theory is about purposive action with different means: the assumption is that all people, regardless of their socioeconomic position, pursue good health, but that they differ in their means to achieve this goal.⁴⁹⁶

This may well be true to some extent, but as others have argued 'differences in means among purposive agents do not account for all the [...] ways that socioeconomic status causes health'. For example, the fact that car drivers with a lower level of education less often put on their seat belts cannot be explained by differences in access to the 'resource' of seat belts, because all cars nowadays have seat belts.⁴⁹⁶ (p. 72)

The suggestion has thus been made to expand the fundamental causes theory to include several other mechanisms: (i) 'Spill-overs within socioeconomic groups.' Even if they do not have the intention to pursue good health as an individual, people with a higher socioeconomic position will still benefit from the purposive action of other individuals in the same group, for example, because their neighbours campaign for a safer neighbourhood. (ii) Different preferences for health. People with a higher socioeconomic position 'may exhibit a stronger and more consistent preference for future good health than others', for example, because of different time horizons or as part of a common culture. (iii) Different treatment by institutions. Schools, medical care facilities and other public institutions may treat people from lower and higher socioeconomic status differently, which may result in different health outcomes, independent from the individual health-directed agency postulated in the original version of the fundamental causes theory.^{xii}

Political economy of health

Nancy Krieger has criticized epidemiology for a lack of attention to theory, and has criticized the available epidemiological theories for a lack of attention to human agency. Of one well-known epidemiological model of the causes of disease, the so-called 'web of causation' which maps all known determinants of a disease as independent factors, she has rhetorically asked: 'where is the spider?', that is, where is the human agency that has created the exposure to all these health determinants?⁴⁹⁷

To some extent this criticism also applies to the theories discussed so far: these are silent on how inequalities in material and non-material resources are generated, and on who benefits from these inequalities. This is even true for the fundamental causes theory, which does not specify how the inequalities in 'flexible resources' come about. If we want to understand why health inequalities are so persistent, we also need to ask whose interests these health inequalities, and the inequalities in their determinants, serve.⁴⁷⁰

Several such theories have been proposed, and **most have an explicitly Marxist perspective emphasizing the role of the economic and political context in generating health inequalities.** For example, in an early attempt to identify the possible links between capitalism and health inequalities, several different pathways were explored and illustrated, such as the link between ‘the imperatives of capital accumulation’ and the need for shift work and use of dangerous chemicals in the workplace, the link between income and housing inequalities, and the ‘contradiction between the pursuit of health and the pursuit of profits.’⁴⁶⁸

Spanish sociologist Vicente Navarro, a Marxist scholar of health and health inequalities, has published widely on the role of economic and political power relations in generating health inequalities. His empirical work has found that political parties with egalitarian ideologies tend to implement redistributive policies, and that policies aimed at reducing social inequalities, such as welfare and labour market policies, reduce (average) infant mortality and increase life expectancy at birth.^{469,498}

In Chapter 5 we will return to Marxist and other viewpoints on the ultimate causes of social inequality.

Key points

- ◆ Research into the explanation of health inequalities has advanced rapidly over the past decades, but recent methodological advances have created uncertainty about the validity of some of the results. This applies particularly to applications of the counterfactual approach to causality (where quasi-experimental studies have tended to cast doubt on the causality of the relationship between socioeconomic position and health) and to advances in mediation analysis (which have cast doubt on the validity of previous estimates of the contribution of specific determinants to health inequalities).
- ◆ At the current state of our knowledge, it seems highly likely that there is a causal effect of education on health, and that the observed association between education and health is therefore partly (but not wholly) due to such a causal effect. It is also likely that there is a causal effect of parental income on children’s health, and while there is no evidence for an effect of modest changes in income on adult health, it is plausible that larger changes in income, particularly at the low end of the income distribution and particularly in countries with lower average incomes, do have an effect on adult health. It is unclear whether there is a causal effect of occupational class on health.
- ◆ Six groups of specific health determinants play a role in the explanation of health inequalities, by linking the ‘social’ to the ‘biological’ throughout an individual’s life-course in mutually reinforcing ways: genetics, childhood environment, material living conditions, social and psychological factors, health-related behaviours, and health care.
- ◆ Nine theories propose a more general explanation of health inequalities and their persistence: ‘social selection’, ‘diffusion of innovations’, ‘cultural capital’, ‘inverse equity’, ‘inverse care law’, ‘neo-materialism’, ‘psychosocial environment’, ‘fundamental causes’ and ‘political economy of health’. While all theories are likely to cover at least part of reality, only the last two emphasize the social origin of health inequalities.